



PE CIVIL-STRUCTURAL EXAM 04

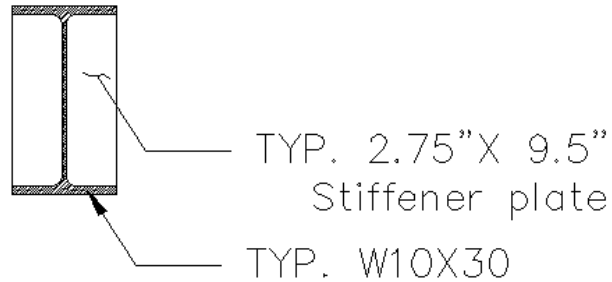
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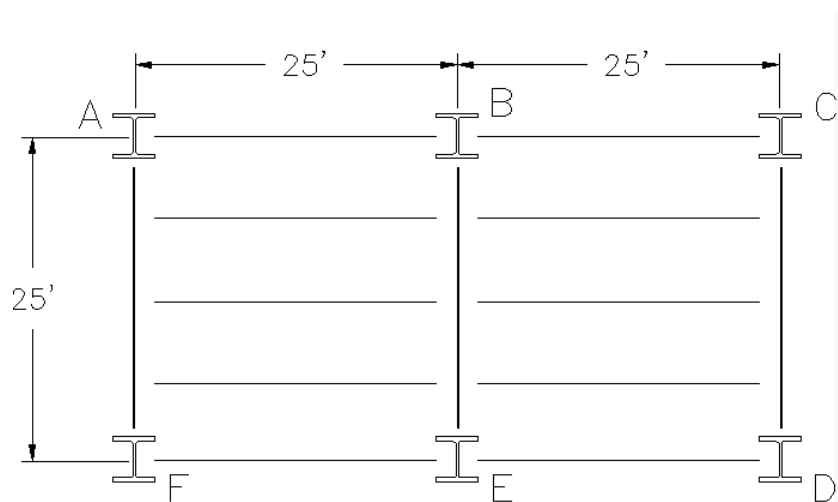
Problems

[Click here for Solutions](#)

1. A W10X30 beam is 20 ft long and has stiffener plates welded on both sides throughout the span, spaced at 4ft center to center. The end bearing stiffeners are 0.5 inches thick on both ends and the inside web stiffeners are 0.25 inches thick. What is the total weight of the beam including the plates?



- (A) 600 lb
(B) 630 lb
(C) 648 lb
(D) 670 lb
2. The partial plan below shows the roof of a single family home. The roof is flat and not intended for occupancy. Calculate the applicable reduction factors and total roof live load acting on column A

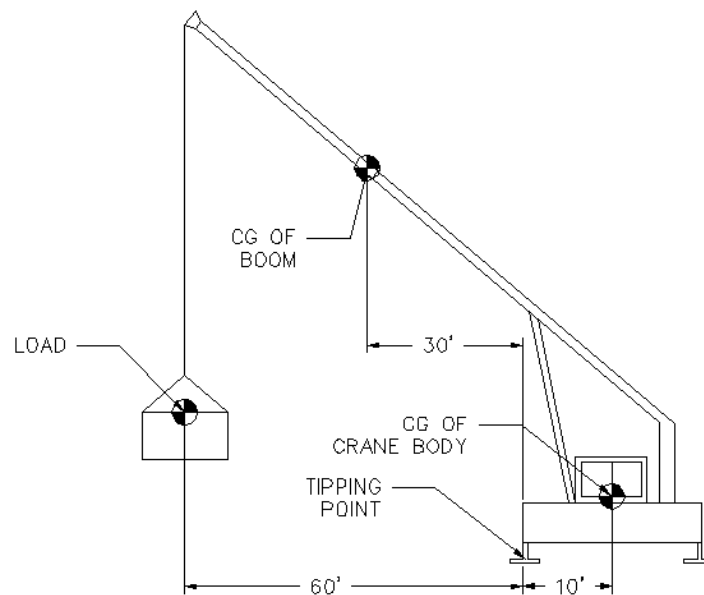


- (A) No roof live load for roof without occupancy
(B) 1562 lb
(C) 3125 lb
(D) 6248 lb

3. The unfactored roof live load acting on a beam is 200 lb/ft . The roof rises 6 inches for every foot of run. The tributary area of the beam is 300ft² . What is the factored reduced live load acting on the beam to be used for strength design?

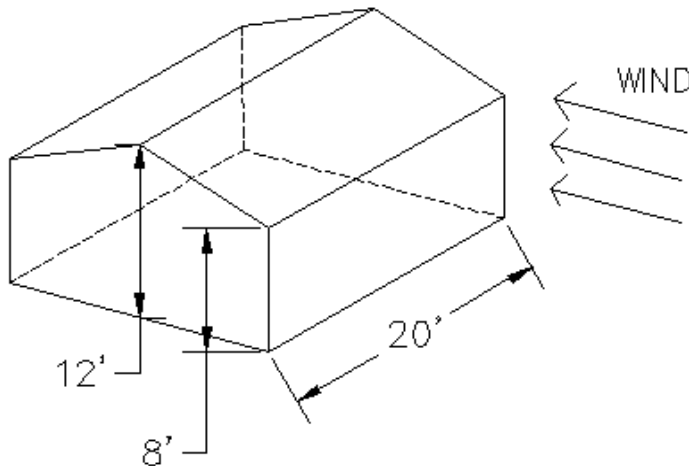
(A) 240 lb
(B) 260 lb/ft
(C) 320 lb/ft
(D) 400 lb/ft

4. A Crane is set at a job site as shown below. The weight of the crane body is 150 kip and the weight of the boom is 15 kip. The crane must lift a load and transport it to a radius of 60 feet from the pick point. What is the maximum tipping load that is allowable to be lifted by the crane. Assume factor of safety 1.5 against overturning.



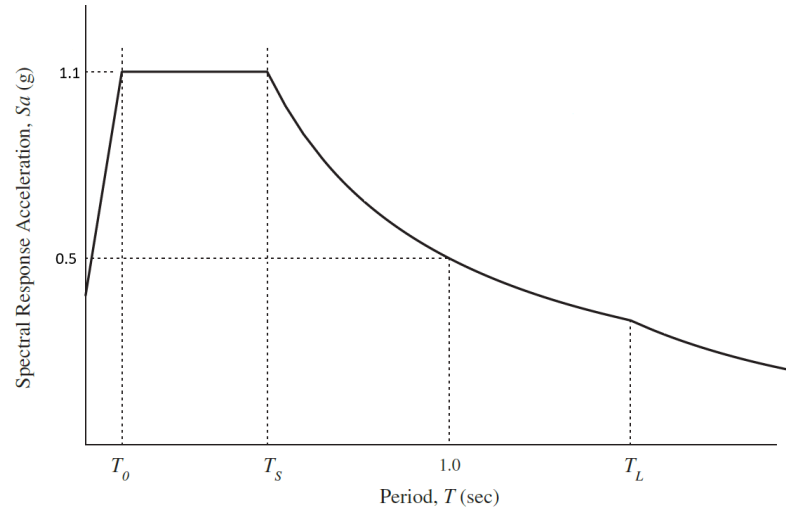
(A) 5 kips
(B) 9 kips
(C) 12 kips
(D) 15 kips

5. A regular shaped enclosed low rise building qualifies for wind analysis using envelope procedure. The calculations yielded a horizontal wind design pressure of 7 psf in the windward direction. What is the total wind load in the projected vertical plane of the windward side as shown below? Follow minimum design requirements per ASCE 7-16.



- (A) 1680 lb
(B) 3200 lb
(C) 3840 lb
(D) 4800 lb
6. A curtain wall is to be replaced in an Exposure C zone. The basic wind speed of the area specified by ASCE 7-16 chapter 26 is 100 mph. The specifications of the material of the curtain wall only list allowable stress values and the strength of the wall against wind loads cannot be checked in accordance with LRFD methodology. What is the basic wind speed to be used in ASD design of the wall?
- (A) 100 mph
(B) 77.5 mph
(C) 85 mph
(D) 90 mph

7. For a site, the horizontal ground motion design response spectrum generated per the procedures of ASCE 7-16 section 11.4.6 is given in the sketch below. The structural design also requires incorporation of vertical ground motions. What is the design vertical response spectral acceleration to be used in the seismic analysis of a building with a vertical period of vibration 0.1 second located at site class D?

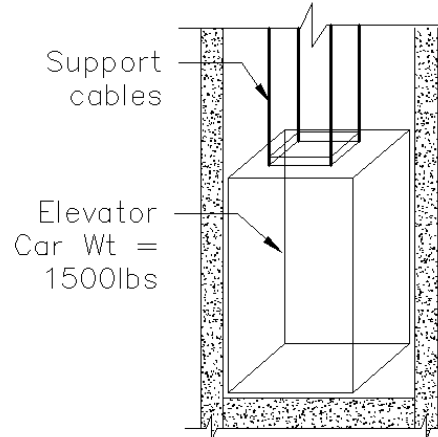


- (A) 1.1g
(B) 1.26g
(C) 1.65g
(D) 2g
8. A 2 story ordinary reinforced concrete moment frame building has an effective seismic weight of 700 kips. The Maximum considered Earthquake short period spectral acceleration is 0.5g. The fundamental period of the structure is 0.2 seconds and the long-period transition period is 4 seconds. Calculate the seismic base shear acting in the structure following equivalent lateral force procedure. Assume risk category II.
- (A) 50 kips
(B) 67 kips
(C) 78 kips
(D) 116 kips
9. Two 25 kips concentrated loads spaced 6 ft apart are moving along the span of a simply supported beam. Calculate the maximum moment acting on the beam. The span of the beam is 25 ft.
- (A) 200 kip*ft
(B) 221 kip*ft
(C) 242 kip*ft
(D) 276 kip*ft

10. During an extreme rainfall event, for a roof in Miami, the hydraulic head for the design flow of the secondary drainage system is 2 inches. When the primary drainage system is blocked, the volume of ponding on the 30ftX30ft roof area is 5,000 gallons . What is the design rain load on the roof?

- (A) 13 psf
- (B) 29psf
- (C) 37psf
- (D) 57psf

11. An elevator car weighs 1500 lbs and is rated for a total passenger weight of 2000 lbs. The elevator is supported by hoisting cables on four corners of the car as shown in the sketch below. What is the minimum safe working load limit of each cable supporting the car? The impact factor to be used for design is 100% .



- (A) 1 kip
- (B) 2 kips
- (C) 3 kips
- (D) 4 kips

12. A reinforced concrete retaining wall has 12ft backfill on one side and 3ft backfill on the other. What is the factor of safety against sliding?

Density of soil, $\gamma = 115$ pcf

Density of concrete, $\gamma_c = 150$ pcf

Active earth pressure coefficient

$= 0.33$

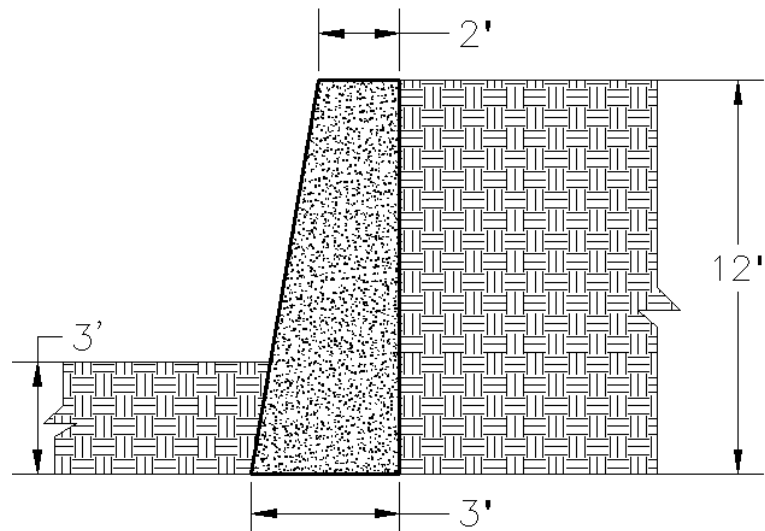
Passive earth pressure

coefficient $= 3$

Coefficient of friction between

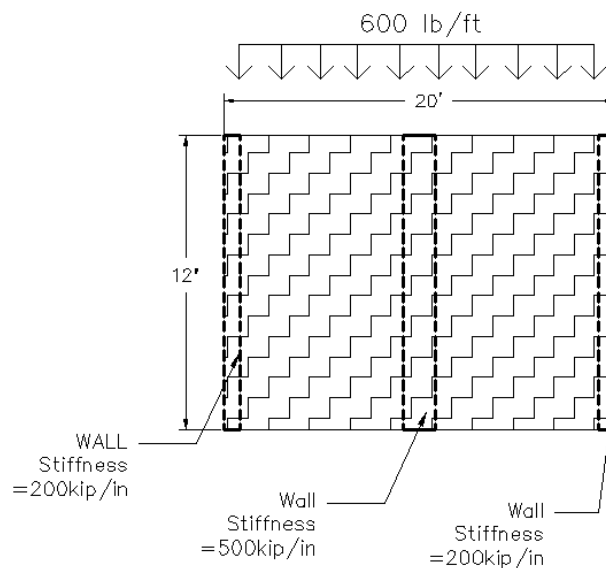
concrete and

soil $= 0.4$



- (A) 0.9
(B) 1
(C) 1.2
(D) 1.5

13. The plan shown below is a light gauge steel deck floor and is supported by three walls underneath. The stiffness of the walls supporting the floor is indicated in the figure. A lateral load of 600 lb/ft is applied on the diaphragm. What is the shear load per unit length acting on the central wall?



- (A) 500 lb/ft
(B) 600 lb/ft
(C) 666.66 lb/ft
(D) 700 lb/ft

14. The maximum bending moments on a roof beam due to different load cases per ASCE 7-16 are as follows:

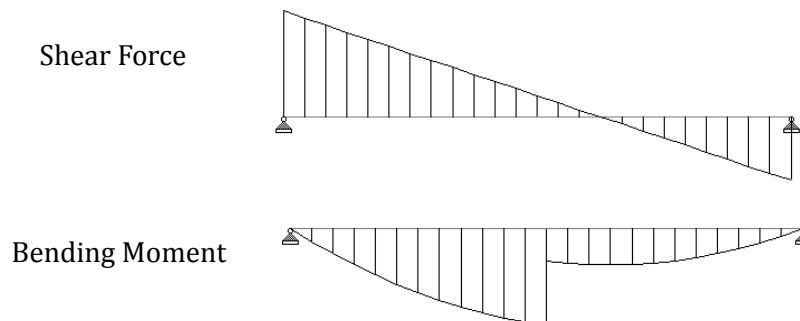
- a. Loads from slab and parapet wall = 40 kip*ft
- b. Load from a water tank = 75 kip*ft
- c. Roof live load = 5 kip*ft
- d. Rain load = 10 kip*ft
- e. Snow load = 8 kip*ft
- f. Vertical seismic load = 5 kip*ft
- g. Horizontal seismic load = 25 kip*ft

What is the governing moment for allowable stress design?

- (A) 112 kip*ft
- (B) 125 kip*ft
- (C) 136 kip*ft
- (D) 137 kip*ft

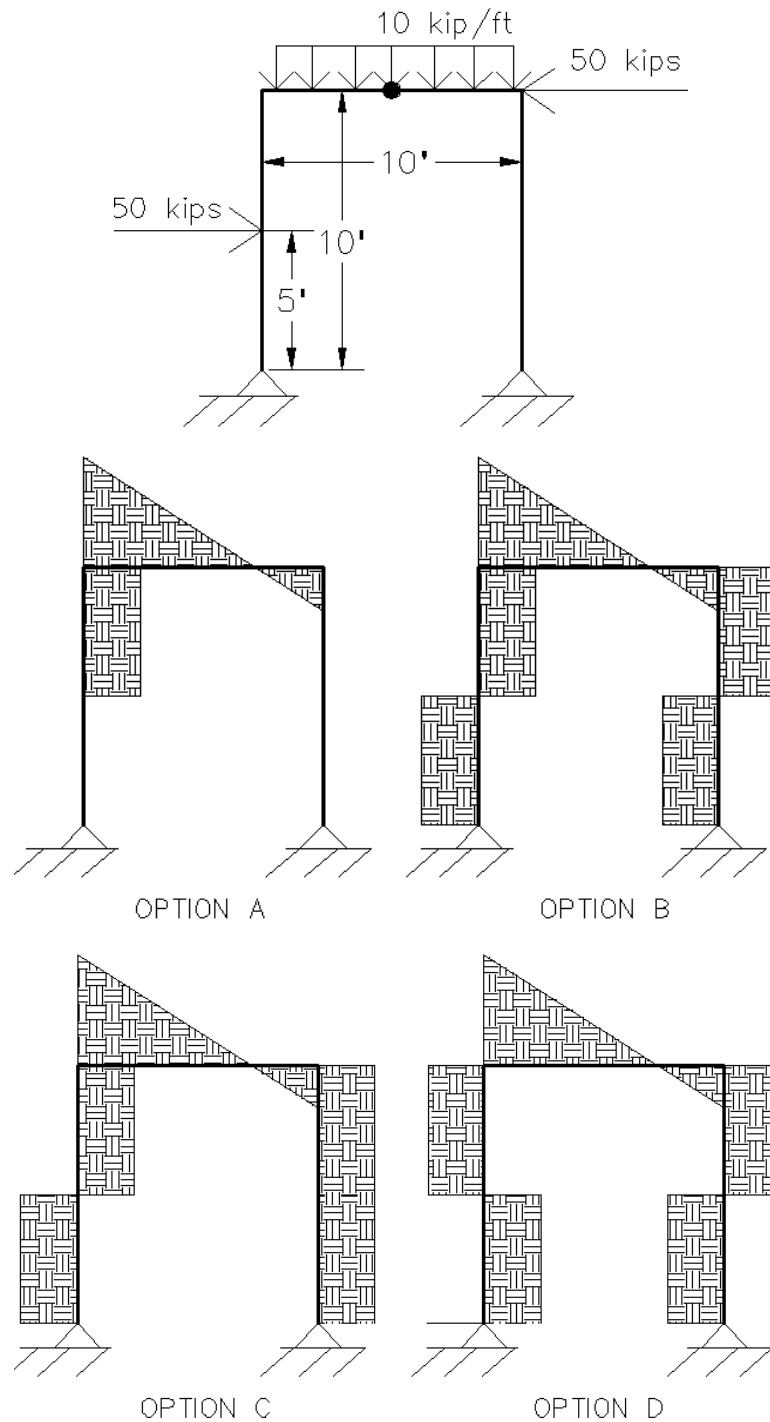
15. For a simply supported beam, the shear force diagram and the bending moment diagram are shown below. Which of the following best describes the loading condition in the beam?

Moment diagram is drawn on the tension side of the beam.



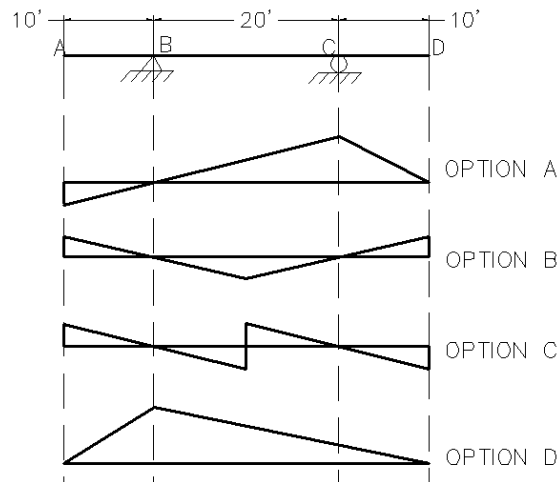
- (A) Uniformly distributed load with a concentrated load at the midspan
- (B) Uniformly distributed load with anti-clockwise moment at the midspan
- (C) Uniformly distributed load with clockwise moment at the midspan
- (D) Uniformly varying load with a couple at the midspan

16. The frame shown below has a hinge at the center of the beam. Select the correct shear force diagram.



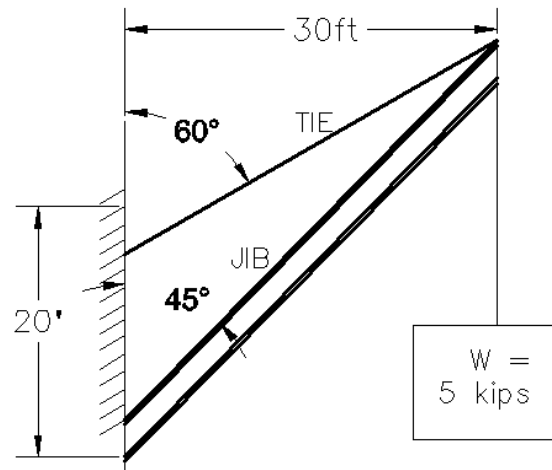
- (A) Option A
- (B) Option B
- (C) Option C
- (D) Option D

17. For the beam shown below, select the correct influence line diagram for maximum shear at midspan, when a concentrated load passes from point A to point D.

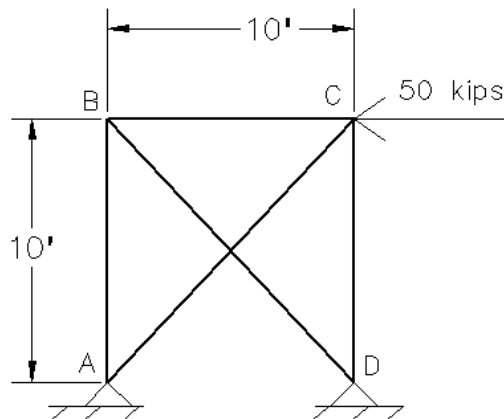


- (A) Option A
 (B) Option B
 (C) Option C
 (D) Option D
18. A cable is suspended between two points and is subjected to a uniformly distributed load of 200 lb/ft. The cable profile is parabolic. Which point in the cable is subjected to the lowest tension?
- (A) Support points
 (B) Mid span at lowest sag point
 (C) Quarter spans
 (D) Cannot be determined

19. The configuration of a maintenance unit crane is as shown below. The pick weight is 5 kips. Both the tie and the jib are rigidly connected to a building wall. What is the axial force acting on the tie?



- (A) 8.66 kips
(B) 11.23 kips
(C) 13.66 kips
(D) 16.73 kips
20. The pin connected frame shown below has a wind load of 50 kips acting on the top. The frame is braced with two threaded tie rods AC and BD forming an X bracing. Calculate the maximum axial load on the braces.

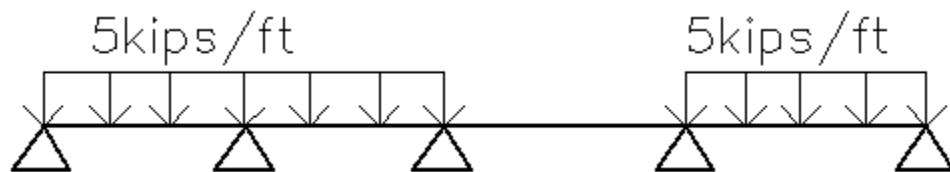


- (A) 35.35 kips
(B) 50 kips
(C) 70.7 kips
(D) 89 kips

21. Which of the following is not true about shear failure in beams and columns?

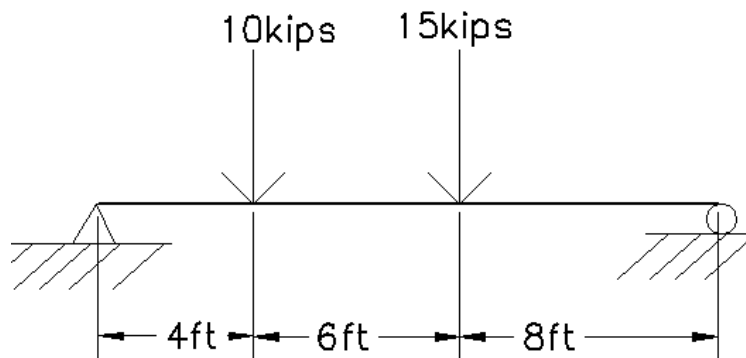
- (A) Short beams and columns are more susceptible to shear failure,
- (B) In short beams, the shear stress is more concentrated making it more susceptible to fail in shear before it develops significant flexural cracks,
- (C) Shear failure may occur in over reinforced sections with high amount of steel,
- (D) The critical value of one way shear in a beam is always at a distance 'd' i.e. effective depth of beam from the face of the support.

22. A continuous beam system is shown below. Each of the four spans are equal in length and = 10ft. What is the shear force at a distance 13 ft from the exterior left support?



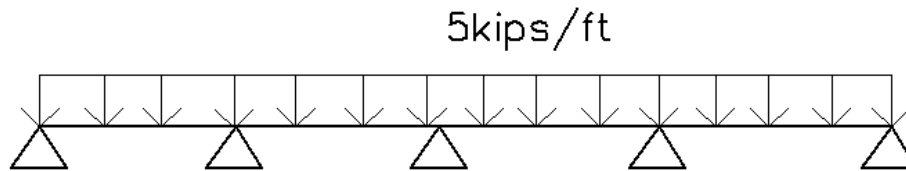
- (A) 10.35 kips
- (B) 15.15 kips
- (C) 20.3 kips
- (D) 30.15 kips

23. A simply supported beam subjected to two point loads is shown below. What is the magnitude of equivalent uniformly distributed load that produces the same amount of maximum shear in the beam?



- (A) 1506 lbs/ft
- (B) 1604 lbs/ft
- (C) 2500 lbs/ft
- (D) 2666 lbs/ft

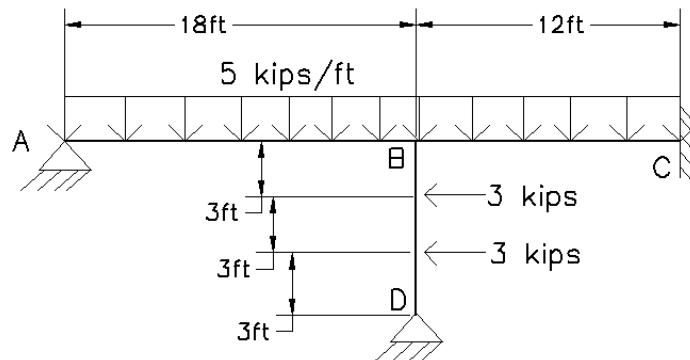
24. The continuous beam system shown below has a uniform distributed load of 5 kips/ft throughout every span. Total length is 40 ft and all four spans are 10 ft in length. W 10 X45 section is used. What is the maximum elastic flexural stress acting on the beam about the strong axis?



- (A) 7 ksi
 (B) 9 ksi
 (C) 11.5 ksi
 (D) 13 ksi
25. Which of the following is not true about point of contraflexure and point of inflection?

- (A) Point of inflection is the point where the bending moment changes from convex to concave or vice versa,
 (B) Point of contraflexure is a point of zero bending moment,
 (C) The shape of elastic curve changes direction at the point of inflection,
 (D) Shear force is either minimum or maximum at the point of inflection.

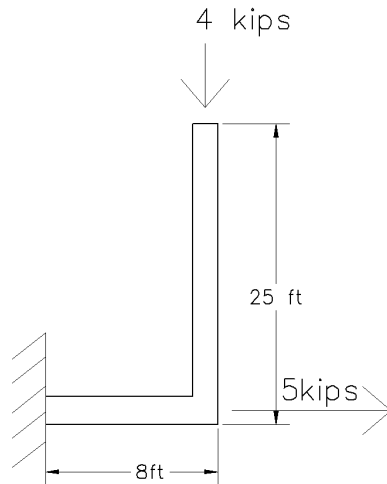
26. For the framing configuration below, what is the reaction moment at point C? The material for the whole framing system is homogeneous. Moment of inertia of beam ABC = 700 in^4 and Moment of inertia for column DB = 1100 in^4 .



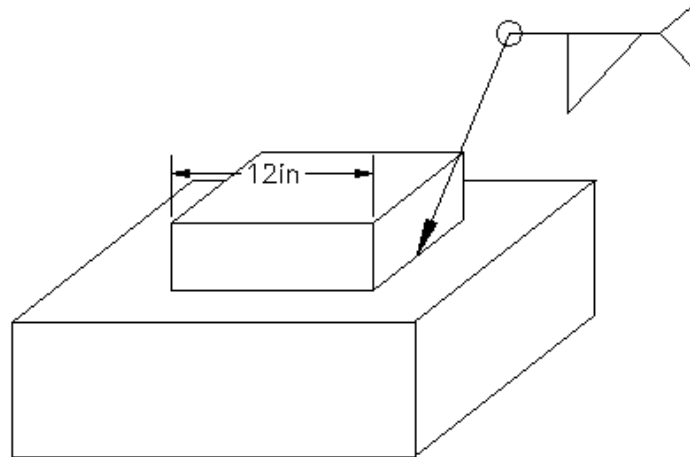
- (A) 23.2 kip*ft
 (B) 31.43 kip*ft
 (C) 38.24 kip*ft
 (D) 55.8 kip*ft

27. A circular pipe bend is shown below. A 4 kip load acts at the top and another force of 5 kip is exerted on the bend as shown below. The pipe's outer diameter is 7 inches and inner diameter is 6 in. What is the maximum normal stress acting in the pipe section?

- (A) 12.32 ksi
(B) 17.56 ksi
(C) 21.23 ksi
(D) 25.26 ksi

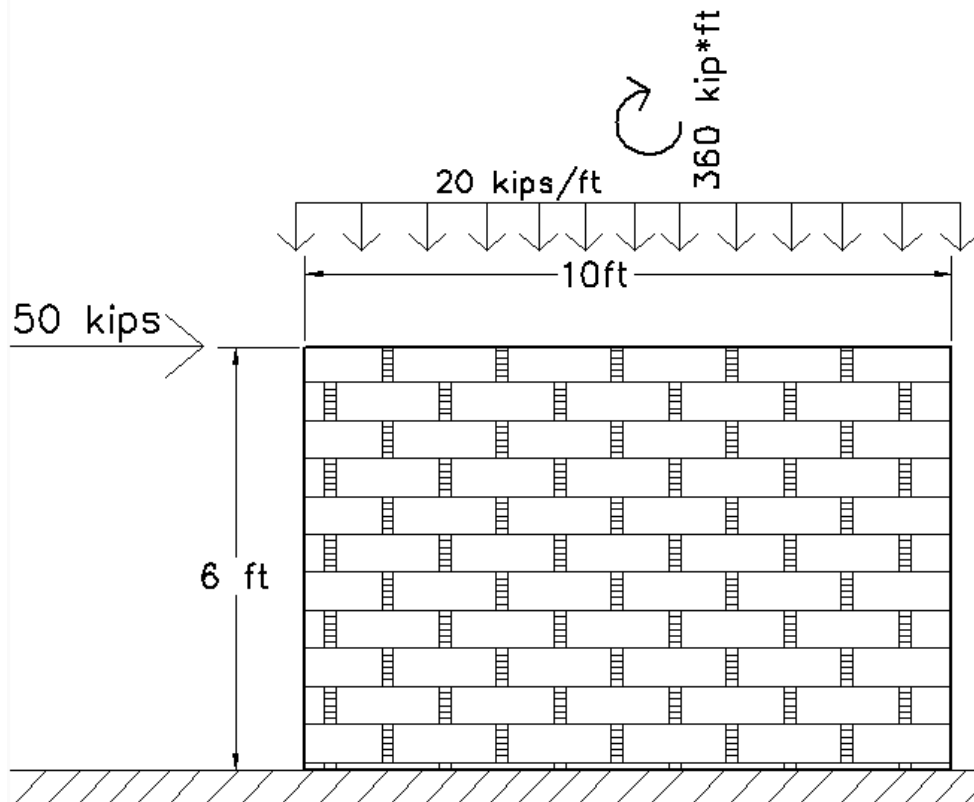


28. A 12inX12 in steel box is welded on top of a base with all round fillet welds as shown below. The net tension load acting on the box is 100 kips and the net horizontal shear acting on the box is 150 kips. What is the resultant load acting on the welds due to combined shear and tension?



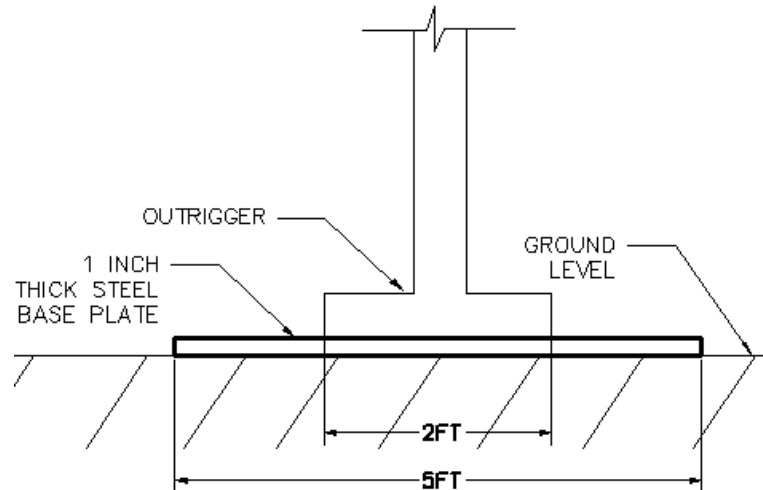
- (A) 2.08 kip/in
(B) 3.125 kip/in
(C) 3.75 kips/in
(D) 5.2 kips/in

29. A masonry wall is subjected to distributed load, a moment and a lateral load at the top as shown in the figure below. What is the maximum tensile stress acting on the masonry wall? The thickness of the wall is 6 inches.

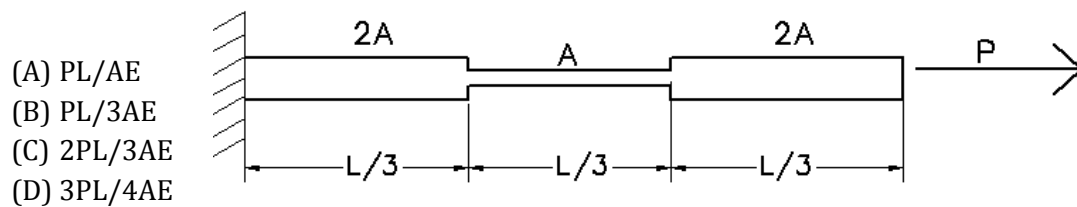


- (A) 0.27 ksi
(B) 0.55 ksi
(C) 0.75 ksi
(D) 0.88 ksi
30. A timber joist has a span 12 ft and is not supporting any ceiling. The joist is subjected to combined dead and live loading. What is the maximum allowable deflection for the joist?
- (A) 0.4 in
(B) 0.8 in
(C) 1 in
(D) 1.2 in

31. An outrigger from a mobile crane is laid on a 5ftX5ft X1in steel base plate. The foot of the outrigger is 2ftX2ft. The maximum concentric axial thrust from the outrigger is 50 kips. What is the maximum deflection of the tip of the steel base plate? $E = 29000$ ksi

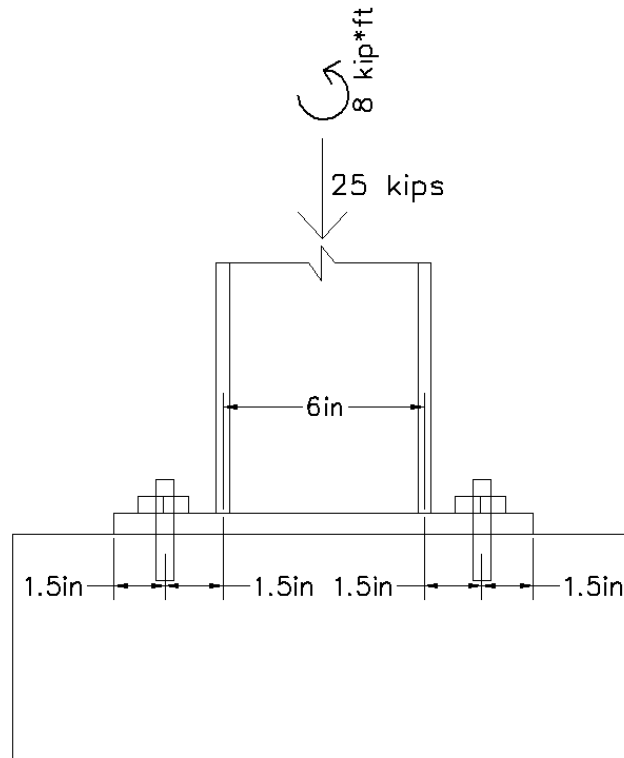


- (A) 0.015 in
(B) 0.03 in
(C) 0.05 in
(D) 0.075 in
32. A beam is subjected to axial load as shown. The cross section areas of the beam vary at different spans as shown below. What is the total deflection of the beam? Modulus of Elasticity is uniform.

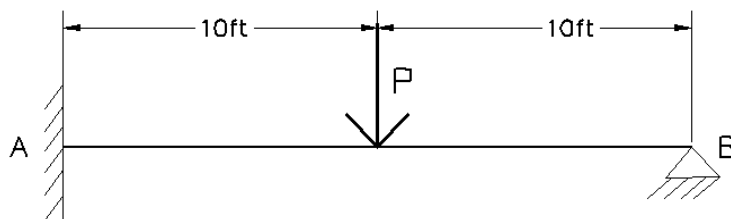


- (A) PL/AE
(B) $PL/3AE$
(C) $2PL/3AE$
(D) $3PL/4AE$

33. The figure below shows a column base plate. The square base plate is 12in X 12 in. Assuming the centroid of the compressive force in the base plate is at the flange of the column, what is the maximum bearing pressure exerted by the loads shown below?



- (A) 0.25 ksi
(B) 0.375 ksi
(C) 0.5 ksi
(D) 0.75 ksi
34. A propped cantilever beam is shown below. The plastic moment capacity of the beam section is 100 kip*ft. What is the value of the collapse load P when the beam is analyzed via plastic analysis?

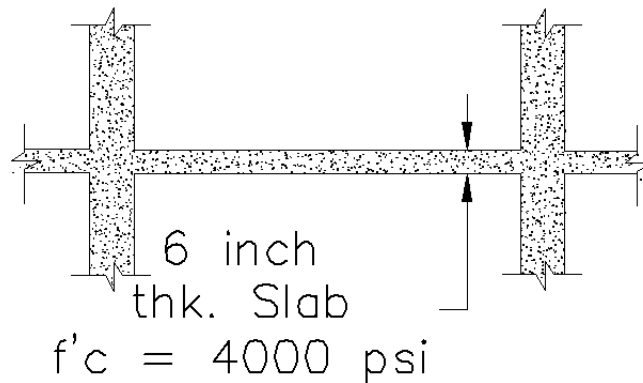


- (A) 28 kip
(B) 30 kip
(C) 32 kip
(D) 34 kip

35. The statement of special inspections that is to be submitted to the building officials is prepared by

- (A) A firm hired by the owner to serve as the engineer of record of the project,
- (B) The contractor's team coordinating all construction permits ,
- (C) The special inspection agency hired for the project who is independent of the contractor,
- (D) The owner's technical personnel who oversee overall construction scheduling and official permits.

36. For a 6 inch thick concrete slab shown below, #5 rebars are spaced at 3 inch c/c at the negative moment zones. What is the maximum aggregate size that can be approved by the special inspector for the pour?



- (A) 1.25 inch
- (B) 1.5 inch
- (C) 1.75 inch
- (D) 2 inch

37. Which of the following is not the purpose of construction submittals?

- (A) To verify overall compliance with design intent and specifications,
- (B) To facilitate record keeping, communication and coordination between different parties,
- (C) To serve as a means of quality control and risk reduction during project implementation,
- (D) To assist the owners with budgeting and financial planning during subsequent phases of the project.

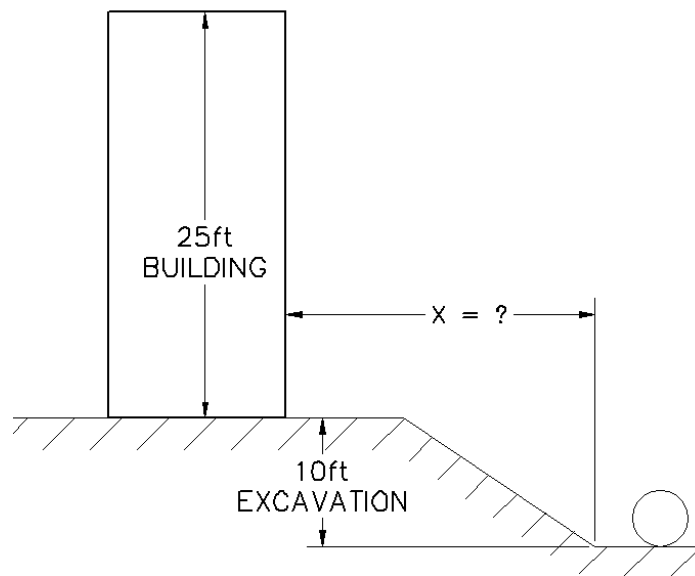
38. What is the most important factor that governs the right time to remove the formwork under beams and slabs?

- (A) The number of days that the member has been cured and the compound used to cure,
- (B) The ambient temperature during curing and construction,
- (C) The compressive strength that the concrete has gained before removal,
- (D) The depth of beams and slabs that are cast.

39. For a 30 feet high, two story wooden scaffold supporting a uniform load of 50 psf, what is the maximum longitudinal and transverse spacing of a 4 in X 4 in wooden pole?

- (A) 6 ft X 5 ft
- (B) 8 ft X 5 ft
- (C) 10 ft X 5 ft
- (D) 12 ft X 5ft

- 40.** Utility lines are to be laid next to a 25 feet tall building at a depth of 10 feet below grade. An undisturbed perimeter of at least half the height of the building has to be maintained around the building. What is the minimum horizontal distance “X” from the face of the building to the bottom of the excavated slope following OSHA’s excavation safety regulations? The soil around the building is Type C per OSHA classifications.



- (A) 18 ft
 (B) 22.5 ft
 (C) 27.5 ft
 (D) 30.75 ft
- 41.** Classify the following inorganic soil sample according to USCS.
 Percentage passing through 0.075 mm sieve = 11%
 Percentage passing through 4.75 mm sieve = 70%
 Liquid Limit = 46 Plastic Limit = 36 $D_{10} = 0.12$ $D_{60} = 0.8$
 Determine the USCS classification of the soil sample.

- (A) SM
 (B) SC
 (C) SW-SM
 (D) SW-SC

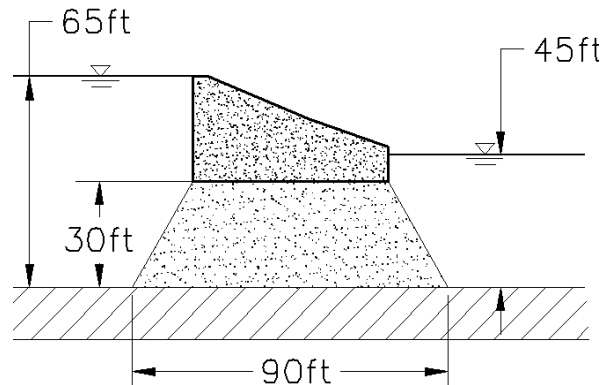
42. Standard proctor test is conducted on a soil sample and the results were as follows:

Mass of sample filling standard mold(lbf)	Water content
4.11	8
4.22	9
4.35	10
4.31	11
4.28	12

The sample tested at site had a total unit weight of 120 pcf and dry unit weight of 112.7 pcf .
What is the estimated relative compaction?

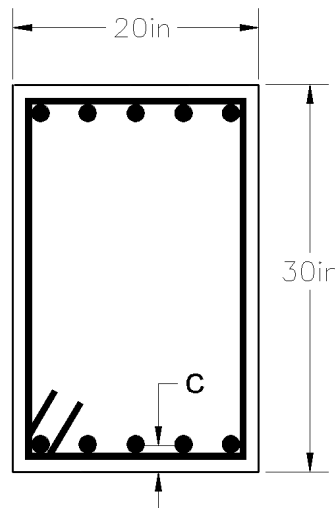
- (A) 81%
- (B) 85%
- (C) 91%
- (D) 95%

43. An embankment is made with impervious material on top of permeable fill. The permeable fill is 30 ft tall, 90 ft in length and has a permeability of 0.004 ft/min. The headwater and tailwater elevations are shown below. What is the discharge through the permeable layer?
The width of the embankment is 300 ft.



- (A) 2 ft³ /min
- (B) 4 ft³ /min
- (C) 6 ft³ /min
- (D) 8 ft³ /min

44. The section of a precast reinforced concrete beam that is manufactured under controlled conditions is shown below. The beam is a part of an open parking lot in a rainy area. What is the minimum distance “c” shown in the figure below? The main bars are number 8 bars and the stirrups are number 4 bars.

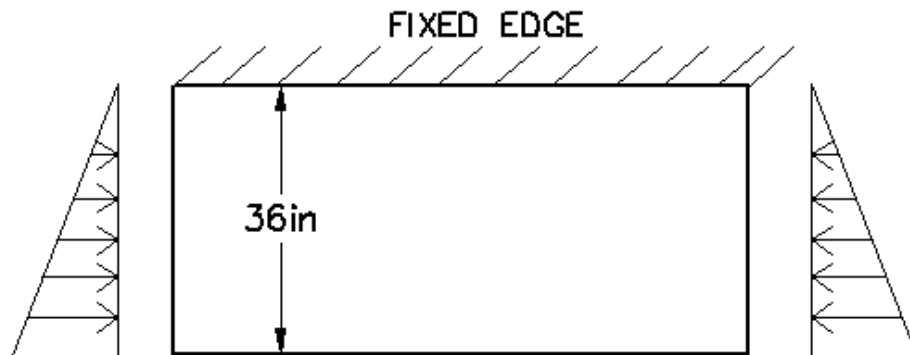


- (A) 2 in
(B) 2.5 in
(C) 3 in
(D) 3.5 in
45. The design strength of concrete during initial prestress is 5000 psi and the 28 day design strength is 6500 psi. What is the allowable compressive stress at the midspan of the uncracked section due to prestressing plus a service live load that causes significant time dependent deflections per ACI?
- (A) 2250 psi
(B) 2925 psi
(C) 3000 psi
(D) 3900 psi

46. Which of the following is not true about strain hardening of steel?

- (A) Strain hardening occurs when material is strained beyond the yield point.
- (B) Strain hardening increases the hardness of the material but reduces the ductility of the and yield strength of the material.
- (C) Strain hardening is used for steel used in Pipelines, storage tanks and pressure vessels.
- (D) Certain plastics like steel also exhibit strain hardening capabilities.

47. A plate is rigidly welded at the top along the length as shown below and loaded along the thickness with a uniformly varying load. The thickness of the plate is 1 in and $E = 29000000$ psi. What is the nominal elastic buckling stress? The yield strength of the plate is 50 ksi.



- (A) 32.5 ksi
- (B) 42 ksi
- (C) 48.5 ksi
- (D) 50 ksi

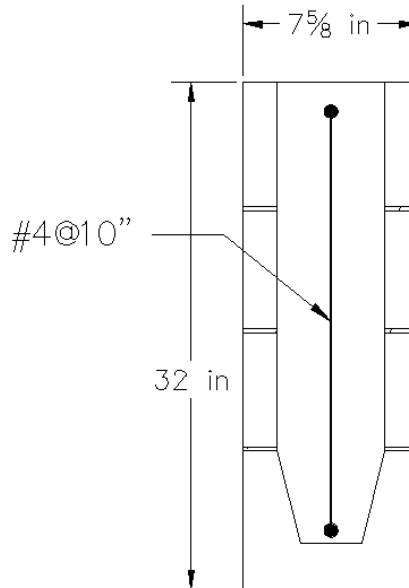
48. A threaded rod is pretensioned to 102 kips during installation. The slip critical connection is in double shear and the surface can be classified as class A. The bolt holes are standard size. What is the allowable shear strength of the connection? There is only one shim plate between the connected parts.

- (A) 20 kips
- (B) 22 kips
- (C) 40 kips
- (D) 44 kips

49. A 4X6 Southern Pine No. 2 is to be used as a tension member. What is the ASD adjusted value of tension for construction loads? Assume 95°F temperature and moisture content of 20 %. Take $C_i = 0.9$.

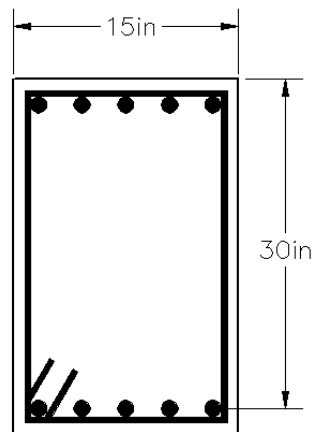
- (A) 600 psi
- (B) 675 psi
- (C) 750 psi
- (D) 900 psi

50. The section of a fully grouted masonry beam is shown below. Grade 60 #4 bars @ 10 inches center to center are used as shear reinforcement. Assuming that the shear force is entirely resisted by the steel reinforcement, what is the maximum allowable shear force?



- (A) 8 kips
- (B) 10 kips
- (C) 12 kips
- (D) 14 kips

51. A reinforced concrete beam is designed and approved to be constructed as shown in the section below. What is the minimum tolerable dimension of the effective depth 'd' while placing the rebars at the construction site per ACI 318-14?



- (A) 29 in
 (B) 29.375 in
 (C) 29.5 in
 (D) 29.75 in
52. For the bolt shown below, which of the following material tests are required to comply with ASTM specs?

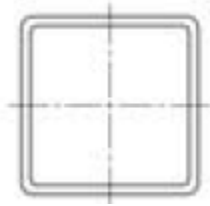


- I) Tensile strength test
 II) Proof load test
 III) Coating test

- (A) I and II
 (B) II only
 (C) I, II and III
 (D) I only

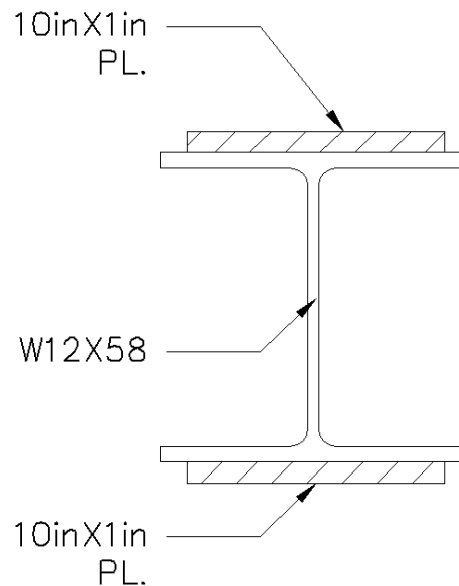
53. A simply supported HSS steel beam has a span of 30 ft and is used as a floor member. The beam is subjected to a service live load of 5 kip/ft and a service dead load of 1 kip/ft and bends about the major axis. What is the lightest section that will satisfy the IBC 2018 serviceability requirements?

$E = 29000 \text{ ksi}$



- (A) HSS 20 X 20 X 1/2
- (B) HSS 20 X 20 X 5/8
- (C) HSS 20 X 20 X 3/4
- (D) HSS 22 X 22 X 3/4

54. 10 inch X 1 inch cover plates are welded effectively to the top and bottom flanges of a W12X58 beam section as shown below. What is the nominal plastic moment of resistance of the section about the major axis? $F_y = 50 \text{ ksi}$ for both plates and rolled section.



- (A) 850 kip*ft
- (B) 910 kip*ft
- (C) 980 kip*ft
- (D) 1055 kip*ft

55. A steel HSS gantry beam is subjected to the following loads:

Axial compression = 20 kips

Strong axis moment = 220 kip*in

Shear = 4 kips

Torsion = 45 kip*in

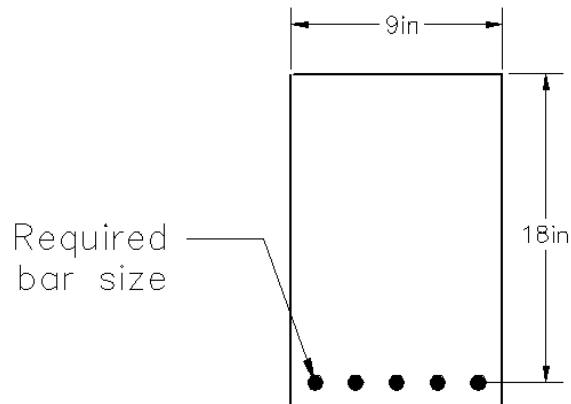
If the allowable moment strength = 500 kip*in, allowable shear strength = 20 kips and the allowable torsional strength is 200 kip*in , what is the minimum required axial strength of the beam?

- (A) 50 kips
- (B) 70 kips
- (C) 100 kips
- (D) 130 kips

56. What is the minimum required section size of a DFL select structural grade lumber, used as a joist that is subjected to a live load moment of 980 lb*ft? Assume normal moisture and temperature conditions. C_i and $C_L = 1$. (Sizes shown are final dressed dimensions)

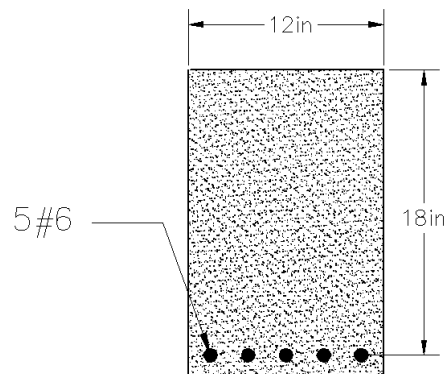
- (A) 2X3
- (B) 2X4
- (C) 2X5
- (D) 2X6

57. Concrete beam section shown below has to resist factored moment of 100 kip*ft. The compressive strength of the concrete is $f'_c = 4$ ksi and Gr. 60 rebars are to be used as flexural reinforcements. Determine the smallest bar size that satisfies the design requirement if five bars are to be used? Assume section satisfies minimum and maximum rebar requirements for all bar sizes and section is tension controlled.



- (A) #4
- (B) #5
- (C) #6
- (D) #7

58. The factored shear to be resisted by the reinforced concrete section shown below is 51 kips. If two legged #3 stirrups are used, what is the maximum spacing of the stirrups required to satisfy the design requirements? Normal weight concrete with compressive strength $f'_c = 4$ ksi, all rebars are Grade 60.



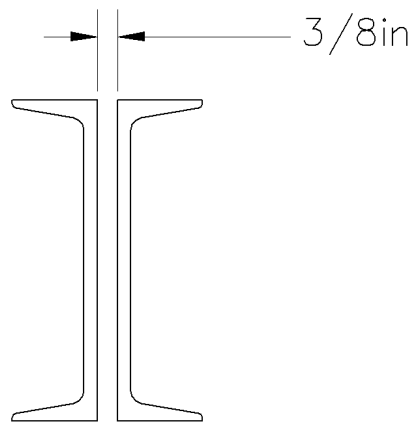
- (A) 6 in
- (B) 9 in
- (C) 12 in
- (D) 10 in

59. An unbraced W8X40 column is subjected to an axial load of 100 kips and the effective length in both directions is 12 ft. The column is also subjected to a strong axis flexure of 25 kip*ft and a weak axis flexure of 10 kip*ft. The compression flange is continuously braced against lateral torsional buckling. What is the total utilization of the column section?

$F_y = 50$ ksi

- (A) 60.3%
- (B) 73.9%
- (C) 82.8%
- (D) 90.1%

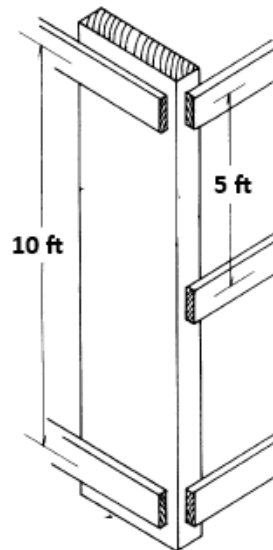
60. A built up column section is made with two channels C 12X30 spaced at 3/8 in back to back as shown below. The effective length in both X and Y direction is 10 ft. What is the nominal critical buckling stress for the column? $F_y = 36$ ksi, $E = 29000$ ksi



- (A) 15 ksi
 - (B) 20 ksi
 - (C) 25 ksi
 - (D) 30 ksi
61. For a masonry column, which of the following is true per requirements set by TMS 402/602-16?
- (A) For a 16in X 16 in masonry column, at least four vertical rebars have to be used and the total area of reinforcement should be atleast 1 in².
 - (B) For a cantilevered column with least radius of gyration 4 in, unbraced height of 17ft is not permitted.
 - (C) Minimum side dimensions of the column should be at least 16in.
 - (D) 28 in X 48 in columns can be tied with #4 bars spaced 28 inches O.C. vertically.

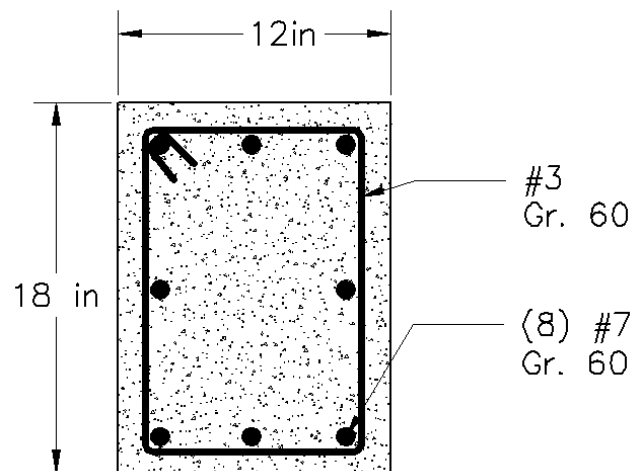
62. A 2X8 DFL 'No. 1 and better', vertical post is braced effectively in both stronger and weaker directions with pin connected bracing planks as shown below. Determine the critical buckling stress F_{CE} using allowable stress design for the section. Assume dry and normal temperature conditions.

$C_i = 1$, $C_T = 1$.



- (A) 339 psi
- (B) 775 psi
- (C) 1257 psi
- (D) 1980 psi

63. For the column section shown below, #3 ties are used as shear reinforcements. What is the maximum permitted tie spacing per ACI 318-14?



- (A) 6 in
- (B) 12 in
- (C) 14 in
- (D) 18 in

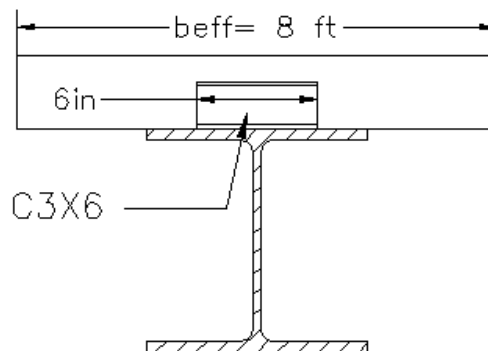
64. In a high wind zone, a five story building is designed with a braced frame system. Which of the following components primarily provides lateral stability in this system?

- (A) Floor diaphragms
- (B) Vertical bracing
- (C) Moment connections
- (D) Foundation anchors

65. A composite steel deck with A992 Gr. 50 steel beams has the steel decking ribs perpendicular to the span of the beam. The concrete used is normal weight 4 ksi. Single 0.5 inch diameter stud per rib is used at the center of the beam. The distance from the edge of the stud shank to the deck web is 2.5 inches. What is the nominal shear strength of the stud if the tensile strength of stud is 65 ksi?

- (A) 6.51 kips
- (B) 7.66 kips
- (C) 9.57 kips
- (D) 11.9 kips

66. A simply supported composite steel beam subjected to uniformly distributed load has Channel C3X6 welded to the top of the steel beam and embedded in a concrete slab with effective width 8 ft. The compressive strength of the concrete is 4 ksi and the modulus of elasticity is 3600 ksi. If the nominal shear to be transferred by the embedded channels is 380 kips, how many channels are required over the full 15 ft span of the beam?

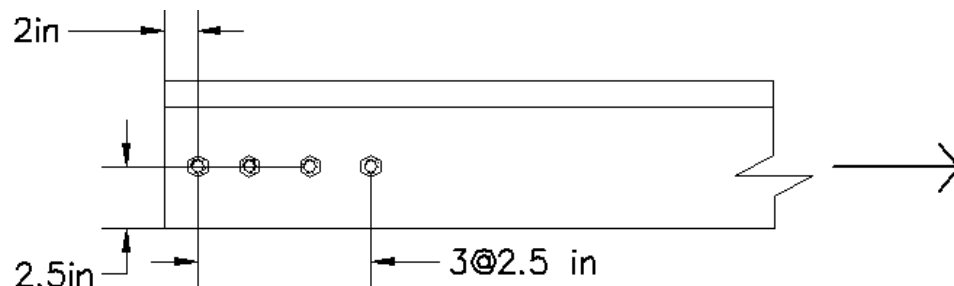


- (A) 2
- (B) 4
- (C) 6
- (D) 8

67. A composite steel deck with 5 inch thick concrete slab and Gr. 50-W18X71 beams, designed for full composite action is unshored before pouring. The factored dead load moment including weight of slab and beam is 75 kip*ft and factored live load moment is 120 kip*ft. What is the factored bending stress at the bottom fibre of the steel beam immediately after pouring?

Section modulus of transformed section measured to the bottom of the section = 185 in³

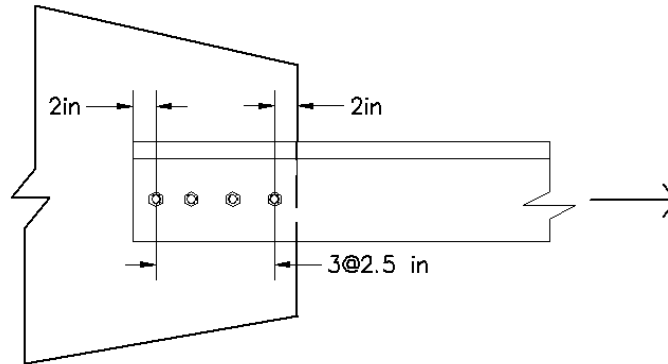
- (A) 4.86 ksi
(B) 7.1 ksi.
(C) 10.3 ksi
(D) 13.4 ksi
68. An angle brace is connected to a gusset plate using a combination of high strength bolts and fillet welds. The design permits sharing of loads between bolts and welds. The maximum tension load acting on the brace is 100 kips. What is the minimum load that the bolts alone must be able to withstand?
- (A) 33 kips
(B) 50 kips
(C) 66 kips
(D) 75 kips
69. An angle section 5X3X1/2 is used as a tension member, connected to a plate with four 0.75 inch diameter bolts as shown below. What is the allowable block shear strength of the angle? Yield strength of angle is 36 ksi and tensile strength is 58 ksi.



- (A) 68 kips
(B) 81 kips
(C) 85 kips
(D) 135 kips

70. For the L 5X3X1/2 connection below, what is the net effective area "Ae" for tensile rupture of the angle as defined by AISC specifications? The bolts used are 0.75 inch diameter. The longer leg of the angle is connected to the plate.

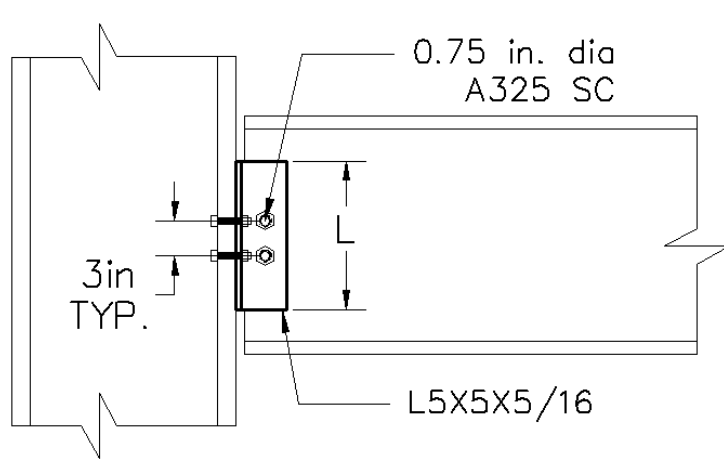
Nominal Hole diameter = 13/16in



- (A) 2.72 in²
(B) 2.98 in²
(C) 3.32 in²
(D) 3.37 in²
71. Which of the following design considerations is most critical to ensure adequate ductility and energy dissipation in steel connections that are subjected to earthquake loading?
- (A) High strength bolts should be used to maximize all connection capacities during cyclic loadings,
(B) The weld sizes in the connections should be increased to reduce stress concentrations during dynamic earthquake loads,
(C) The connection should be designed such that connected parts yield before the fasteners or welds fails,
(D) The number of bolts should be minimized to simplify the seismic load path,

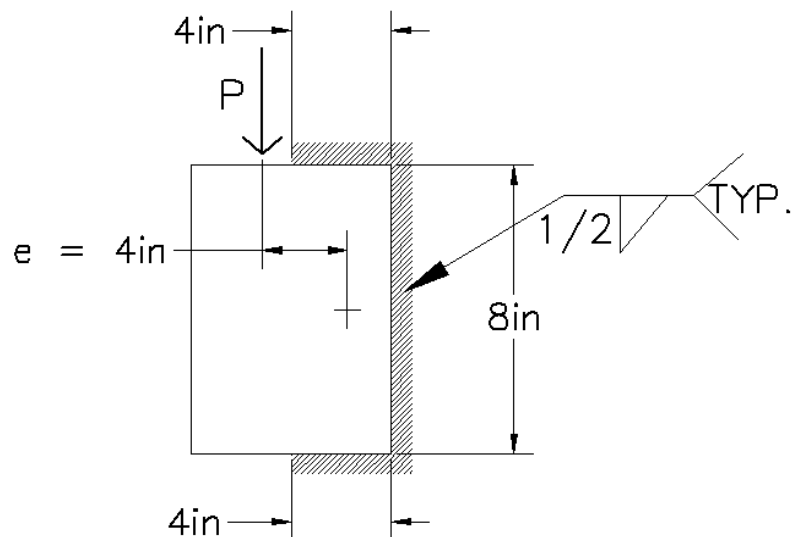
72. A double angle beam to column shear connection is made using 5X5X5/16 angles on both sides of the beam web as shown below (number of bolts shown for illustration of connection type only). The spacing of the bolts will be kept at 3 inches consistent and Length of the angle 'L' varies per number of bolts used such that the top and bottom bolt edge distances are at least 1.25 in. The horizontal edge distance for the bolts will be maintained at 1.375 in. minimum. The connection has to transfer a service dead load reaction of 20 kips and a service live load reaction of 50 kips. Assuming failure of connection is governed by the angle and bolts, what is the number of slip critical (class A) A325 , 3/4 inch diameter bolts

required for the beam web?
 Beams and columns are A992.
 Yield strength of angle = 36 ksi
 Tensile strength = 58 ksi.
 All holes are standard size.



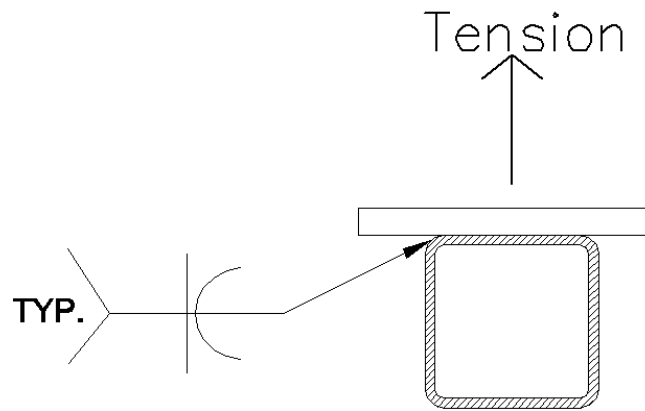
- (A) 2
 (B) 4
 (C) 6
 (D) 8

73. A plate has welds on three sides as shown below. Using instantaneous center of rotation method, what is the maximum allowable load that 'P' that the plate can carry, assuming that the welds govern the failure of connection.
 $F_{exx} = 70$ ksi

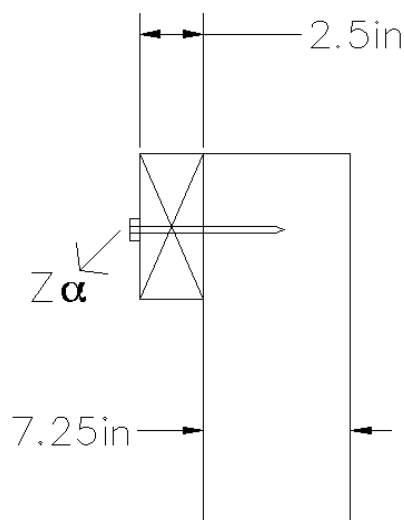


- (A) 51.3 kips
 (B) 66.4 kips
 (C) 78.4 kips
 (D) 91.2 kips

74. HSS 8X8X1/4 section is welded to a top plate with flare bevel welds flush to the outer wall of the HSS as shown below. The length of the plate along the HSS span is 6 inches. What is the allowable concentric tension load on the plate connection? $F_{exx} = 70$ ksi (SMAW). Assume HSS walls and plate do not govern failure.



- (A) 19.5 kips
(B) 39 kips
(C) 48 kips
(D) 55.3 kips
75. A standard 7 inch long 3/8 inch diameter hex lag screw with minimum thread length is used to fasten two Douglas Fir-Larch members as shown below. The fastener is subjected to a loading at an angle 45° . What is the allowable load in the fastener ? Assume all other adjustment factors are equal to 1.



- (A) 140 lbs
(B) 250 lbs
(C) 305 lbs
(D) 325 lbs

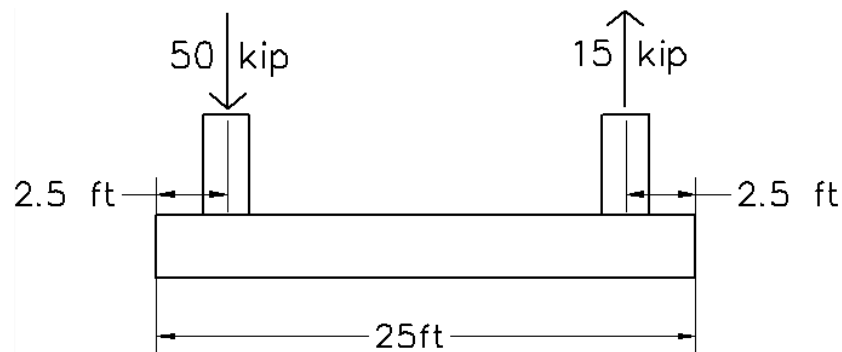
76. The top of a 25ftX50ft concrete mat foundation is 5 ft below the ground level. The dead load acting on the mat due to superstructure is 1500 kips and the uplift due to wind load is 1800 kips. What is the minimum required depth of the mat if a factor of safety of 1.5 is desired against the uplift?

Unit weight of concrete = 145 pcf

Unit weight of soil = 110 pcf

- (A) 2 ft
- (B) 2.75 ft
- (C) 3 ft
- (D) 4 ft

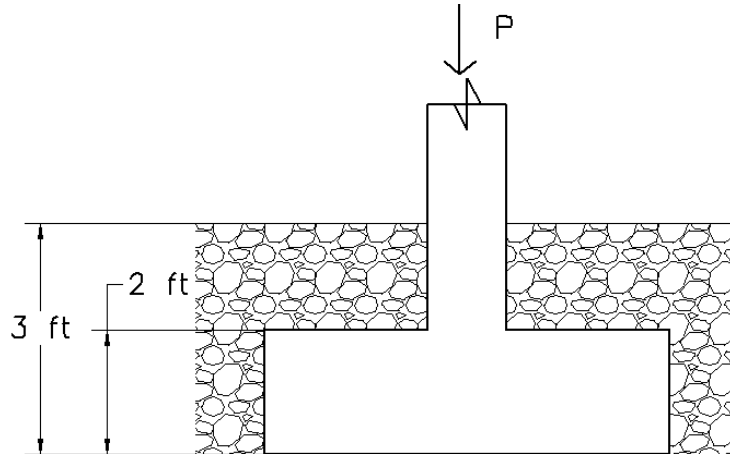
77. A combined footing supports two columns as shown below. The self weight of the footing is 1.8 klf and the width of the footing is 6ft. What is the maximum bearing pressure under the footing?



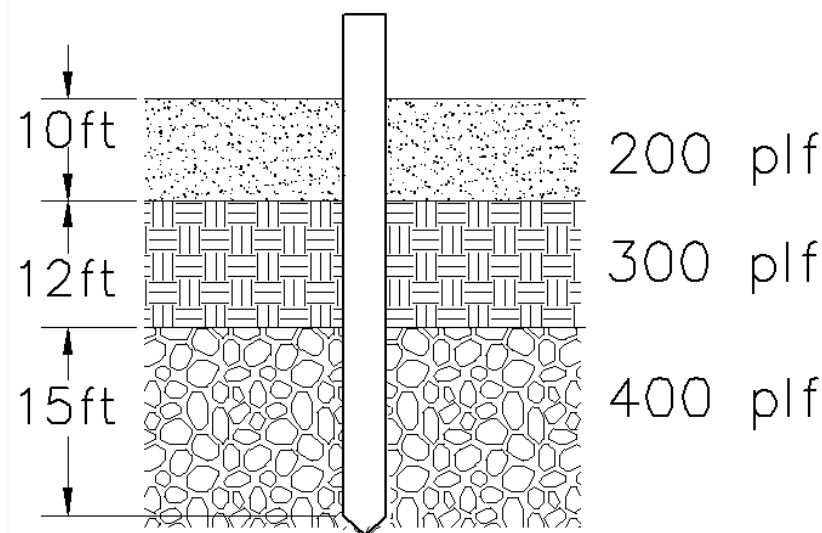
- (A) 0.53 ksf
- (B) 1.55 ksf
- (C) 1.74 ksf
- (D) 2.03 ksf

78. A square footing is located at a site where the bearing layers are classified as sandy gravel and gravel (GW and GP). What is the required width of the footing if a concentric compression of 40 kips acts in the column following IBC provisions?
Unit weight of soil = 115 pcf. Unit weight of concrete = 150 pcf.

- (A) 3.5 ft
(B) 4 ft
(C) 4.5 ft
(D) 5 ft



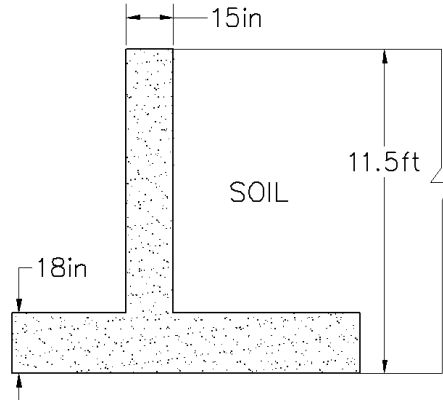
79. A friction pile is driven through different layers of soil strata in a river bed as shown below. The allowable skin friction of each layer per unit length of pile is given next to the layer. The hydraulic specialist determined that the scour depth due to water currents in the river bed is equal to 8 ft. What is the allowable axial load capacity of the piles?



- (A) 10 kips
(B) 11.5 kips
(C) 12 kips
(D) 15 kips

- 80.** A reinforced concrete cantilevered wall is retaining 11.5 ft tall backfill. What is the distance of the critical location for one way shear demand calculation of the stem, from the top of the wall? Effective cover of tension bars at the stem = 3 in. Density of soil = 120 pcf, Unit weight of concrete = 150 pcf, Active earth pressure coefficient = 0.33

- (A) 5.75 ft
- (B) 9 ft
- (C) 10 ft
- (D) 11.5 ft



Solutions

[Click here for Problems](#)

1. Problem

Correct Answer: (B) 630 lb

Explanation:

Length of beam = 20ft

Weight of W10X30 = 30 lb/ft {Ref AISC manual table 1-1}

Beam weight = 30lb/ft* 20ft = 600 lb

Density of steel = 490 lb/ft³

Calculate the weight of stiffener plates

Plate 1 = 490 lb/ft³ * 2.75in * 9.5in * 0.25in = 1.85 lb

Plate 2 = 490 lb/ft³ * 2.75in * 9.5in * 0.5in = 3.7 lb

Spacing of stiffeners = 4ft c/c

Total number of stiffened points = 20ft/4ft+1 = 6. These plates are on both sides of the web.

Thus, there are 2X2 end bearing plates and 4X2 inside web stiffeners.

Total weight = Weight of beam + weight of end plates + weight of inside plates
= 600lb + 2*2*3.7lb+4*2*1.85lb = 629.6 lbs ≈ **630 lb**

2. Problem

Correct Answer: (C) 3125 lbs

Explanation:

Per ASCE 7-16 page 15, Table 4.3-1 minimum live load on roof without occupancy is 20psf.

Per ASCE 7-16 , page 17, section 4.8.2

Check applicable reduction factors,

Tributary area for column A

$A_T = (25\text{ft} \times 25\text{ft}) / 4 = 156.25 \text{ ft}^2$ (less than 200ft²)

For flat roof, pitch factor F = 0

Thus reduction factors R_1 and R_2 are both 1, hence the roof live load cannot be reduced.

Live load to be considered = 20 psf

Total live load acting on column A = 20psf* 156.25 ft² = **3125 lbs**

3. Problem

Correct Answer: (B) 260 lb/ft

Explanation:

Tributary area = 300 ft^2

Per ASCE 7-16, page 17, section 4.8.2, the reduction factors for roof live load are

$$R_1 = 1.2 - 0.001 \cdot A_T \text{ for } 200 \text{ ft}^2 < A_T < 600 \text{ ft}^2$$

$$= 1.2 - 0.001 \cdot 300$$

$$= 0.9$$

Calculate pitch of roof

Pitch = 6 inches/1 feet

Thus, $F = 6$

$$R_2 = 1.2 - 0.05 \cdot F \text{ for } 4 < F < 12$$

$$= 1.2 - 0.05 \cdot 6$$

$$= 0.9$$

Calculate total reduced load $L_R = 200 \text{ lb/ft} \cdot 0.9 \cdot 0.9$

$$= 162 \text{ lb/ft}$$

Factored roof live load for strength design = $1.6 \cdot 162 \text{ lb/ft} = 259.2 \text{ lb/ft} \approx \mathbf{260 \text{ lb/ft}}$

4. Problem

Correct Answer: (B) 9 kips

Explanation:

Sum of moments about the tipping point must be zero.

Resisting moments about tipping point due to self weight of crane body

$$M_R = 150 \text{ kip} \cdot 10 \text{ ft}$$

$$= 1500 \text{ kip} \cdot \text{ft}$$

Overturning moments about tipping point due to load and boom self weight

$$M_O = \text{Load} \cdot 60 \text{ ft} + 15 \text{ kip} \cdot 30 \text{ ft}$$

For equilibrium, including a factor of safety 1.5 against overturning

$$1.5 (\text{Load} \cdot 60 \text{ ft} + 15 \text{ kip} \cdot 30 \text{ ft}) = 1500 \text{ kip} \cdot \text{ft}$$

$$\text{Load} \cdot 60 \text{ ft} = 1500 \text{ kip} \cdot \text{ft} / 1.5 - 450 \text{ kip} \cdot \text{ft}$$

Solve for Tipping load = $9.166 \text{ kips} \approx \mathbf{9 \text{ kips}}$

5. Problem

Correct Answer: (B) 3200 lbs

Explanation:

Per ASCE 7-16 section 28.3.4, minimum design wind loads should not be less than 16 psf for area of walls and 8 psf for area of roofs projected in vertical plane.

Projected roof area in vertical plane

$$= (12\text{ft} - 8\text{ft}) * 20\text{ft} = 80\text{ft}^2$$

Projected wall area in vertical plane

$$= 8\text{ft} * 20\text{ft} = 160\text{ft}^2$$

$$\text{Load for projected roof area} = 80\text{ft}^2 * 8\text{psf} = 640\text{ lbs}$$

$$\text{Load for projected wall area} = 160\text{ft}^2 * 16\text{psf} = 2560\text{ lbs}$$

$$\text{Total load} = 640\text{ lbs} + 2560\text{ lbs} = \mathbf{3200\text{ lbs}}$$

6. Problem

Correct Answer: (B) 77.5 mph

Explanation:

The basic wind speeds given in maps of ASCE 7-16 and IBC 2018 are ultimate design wind speeds to be used in LRFD design load combinations.

To convert it to ASD wind speeds and get ASD design wind pressures, the wind speed has to be scaled down using IBC 2018 equation 16-33, page 378

$$V_{asd} = V \sqrt{0.6}$$

$$V_{asd} = 100\text{mph} \sqrt{0.6}$$

$$V_{asd} = 77.46\text{mph} \approx \mathbf{77.5\text{ mph}}$$

7. Problem

Correct Answer: (B) 1.26g

Explanation:

Given, vertical period of vibration of structure = 0.1s

From the given graph of design horizontal spectrum, design short period spectral acceleration, $S_{DS} = 1.1g$

Back calculate the value of S_{MS} from per ASCE 7-16 section 11.4.5, page 84

$$S_{DS} = (2/3) * S_{MS}$$

$$S_{MS} = (3/2) * S_{DS} = (3/2) * 1.1g = 1.65g$$

Back calculate the value of S_s from equation 11.4-1,

$$S_{MS} = F_a * S_s$$

$F_a = 1$ from table 11.4-1 for site class D and assumed $S_s > 1.5g$

Thus, $S_s = S_{MS} / F_a = 1.65g$ (Assumption $> 1.5g$ OK)

From ASCE 7-16, section 11.9.2, for vertical period of vibration greater than 0.05s and less than 0.15, equation 11.9-3 can be used to calculate MCE_R vertical response spectral acceleration

$$S_{aMv} = 0.8 * C_v * S_{MS}$$

From Table 11.9-1, value of coefficient $C_v = 1.3$ for $S_s = 1$ and $C_v = 1.5$ for $S_s = 2$

Interpolating for $S_s = 1.65$, we get $C_v = 1.43$

$$\text{Then, } S_{aMv} = 0.8 * C_v * S_{MS} = 0.8 * 1.43 * 1.65g = 1.89g$$

From section 11.9.3, the design vertical response spectral acceleration

$$S_{av} = (2/3) * S_{aMv} = 0.667 * 1.89g = \mathbf{1.26g}$$

8. Problem

Correct Answer: (C) 78 kips

Explanation:

From ASCE 7-16 page 5, Table 1.5-2, for risk category II,

Seismic Importance factor $I_e = 1$

From ASCE 7-16 page 91, Table 12.2-1, for ordinary reinforced concrete moment frame,

$R = 3$

Given, $S_{MS} = 0.5g$

$$S_{DS} = (2/3) * S_{MS} = 0.333g$$

From equation 12.8-2, page 101

$$C_s = S_{DS} / (R / I_e) = 0.333 / (3/1) = 0.111$$

Given, effective seismic weight $W = 700$ kips

From equation 12.8-1, Base shear is calculated as

$$V = C_s * W = 0.111 * 700 \text{ kips} = 77.7 \text{ kips} \approx 78 \text{ kips}$$

9. Problem

Correct Answer: (C) 242 kip*ft

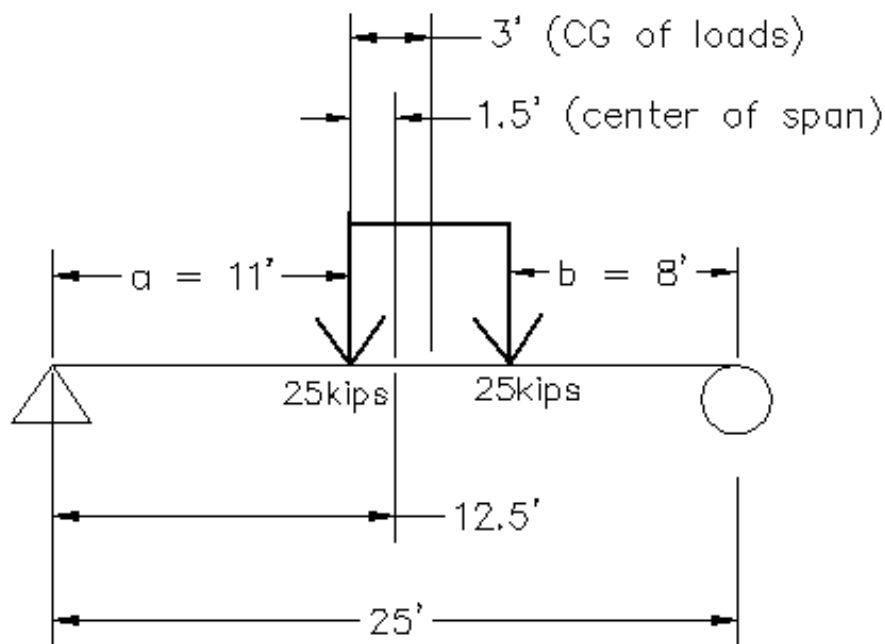
Explanation:

Per NCEES handbook page 232, Section 4.1.2, the absolute maximum moment produced in a beam by a set of n moving loads occurs when the resultant R of the load set and an adjacent load are equidistant from the centerline of the beam.

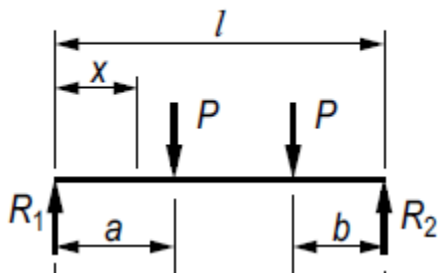
Per the question, two 25 kips loads are moving along the span, spaced 6 ft apart.

CG of loading group i.e. resultant of load is at $6\text{ft}/2 = 3\text{ft}$.

Thus, condition for maximum moment is when the center of the 25 ft long beam i.e. 12.5 ft is between this CG and any one of the axle as shown in the sketch below



Then use the formula from NCEES handbook page 239 (Condition 10)



$R_1 = V_1 (=V_{\max} \text{ when } a < b)$	$= \frac{P}{l}(l - a + b)$
$R_2 = V_2 (=V_{\max} \text{ when } a > b)$	$= \frac{P}{l}(l - b + a)$
$V_x [\text{when } a < x < (l - b)]$	$= \frac{P}{l}(b - a)$
$M_1 (=M_{\max} \text{ when } a > b)$	$= R_1 a$

Here, $P = 25 \text{ kips}$ $a = 11 \text{ ft}$ $b = 8 \text{ ft}$ $l = 25 \text{ ft}$

$R_1 = 25 \text{ kip} / 25 \text{ ft} * (25 \text{ ft} - 11 \text{ ft} + 8 \text{ ft}) = 22 \text{ kip}$

$M_{\max} = R_1 * a = 22 \text{ kip} * 11 \text{ ft} = \mathbf{242 \text{ kip}\cdot\text{ft}}$

10. Problem

Correct Answer: (D) 57 psf

Explanation:

From NCEES PE handbook, page 3

1 gallon = 0.134 ft^3

5000 gallons = $5000 * 0.134 \text{ ft}^3 = 670 \text{ ft}^3$

Depth of water due to blockage, i.e. static head = Volume / Area

$d_s = 670 \text{ ft}^3 / (30 \text{ ft} * 30 \text{ ft})$

$d_s = 0.7444 \text{ ft} = 8.9 \text{ inches}$

Given, Hydraulic head

$d_h = 2 \text{ inches}$

From ASCE 7-16 page 65, equation 8.3-1 or IBC 2018 page 391 equation 16-35

Rain load

$R = 5.2 (d_s + d_h)$

$= 5.2 * (8.9 + 2)$

$= 56.68 \text{ psf} \approx \mathbf{57 \text{ psf}}$

11. Problem

Correct Answer: (B) 2 kips

Explanation:

Weight of cab = 1500 lbs

Weight of passengers = 2000 lbs

Total weight = 1500 lbs + 2000 lbs = 3500 lbs = 3.5 kip

Apply impact factor 100%

Service load with impact factor

$$= 3.5 \text{ kip} * (1 + 100\%)$$

$$= 3.5 \text{ kip} * (1 + 1)$$

$$= 7 \text{ kip}$$

Hence safe working load limit for each supporting cable

$$= 7 \text{ kip} / 4 = 1.75 \text{ kip}$$

Correct answer from options is **2 kips**

12. Problem

Correct Answer: (C) 1.2

Explanation:

Given,

Active earth pressure coefficient, $K_a = 0.33$

Passive earth pressure coefficient, $K_p = 3$

Density of soil, $\gamma = 115 \text{ pcf}$

Density of concrete, $\gamma = 150 \text{ pcf}$

Height of embankment for active pressure = 12ft

Height of embankment for passive pressure = 3ft

Calculate self weight of concrete that resists sliding

$$\text{Area of wall} = 0.5 * 12\text{ft} * (2\text{ft} + 3\text{ft}) = 30\text{ft}^2$$

$$\text{Self weight of wall} = \text{Area} * \text{density} = 30\text{ft}^2 * 150\text{pcf} = 4500\text{lb/ft}$$

Calculate active pressure

$$= \frac{1}{2} * K_a * \gamma * H^2$$

$$= 0.5 * 0.33 * 115\text{pcf} * 12\text{ft} * 12\text{ft}$$

$$= 2732.4 \text{ lb/ft}$$

Calculate resisting passive pressure

$$= \frac{1}{2} * K_p * \gamma * H_p^2$$

$$= 0.5 * 3 * 115 \text{ pcf} * 3 \text{ ft} * 3 \text{ ft}$$

$$= 1552.5 \text{ lb/ft}$$

Calculate resisting force due to friction between concrete and soil

= coefficient of friction * weight of concrete wall

$$= 0.4 * 4500 \text{ lb/ft}$$

$$= 1800 \text{ lb/ft}$$

$$\text{Total sliding force} = 2732.4 \text{ lb/ft}$$

$$\text{Total resisting force} = 1552.5 \text{ lb/ft} + 1800 \text{ lb/ft} = 3352.5 \text{ lb/ft}$$

$$\text{Factor of safety against sliding} = 3352.5 / 2732.4 = \mathbf{1.2}$$

13. Problem

Correct Answer: (A) 500 lb/ft

Explanation:

Light gauge steel deck floor is idealized as a flexible diaphragm.

A flexible diaphragm distributes load according to tributary area and not stiffness

Tributary area for interior wall

$$= 20 \text{ ft} / 2 * 12 \text{ ft}$$

$$= 120 \text{ ft}^2$$

$$\text{Total area of floor} = 20 \text{ ft} * 12 \text{ ft} = 240 \text{ ft}^2$$

Load shared by central wall

$$= (120 \text{ ft}^2 / 240 \text{ ft}^2) * (600 \text{ lb/ft} * 20 \text{ ft})$$

$$= 6000 \text{ lbs}$$

$$\text{Load per unit length of wall} = 6000 \text{ lbs} / 12 \text{ ft} = \mathbf{500 \text{ lb/ft}}$$

14. Problem

Correct Answer: (D) 137 kip*ft

Explanation:

$$D = 40 \text{ kip*ft}$$

$$F = 75 \text{ kip*ft}$$

$$L_r = 5 \text{ kip*ft (does not govern over rain load)}$$

$$S = 8.35 \text{ kip*ft (does not govern over rain load)}$$

$$R = 10 \text{ kip*ft}$$

$$E_h = 25 \text{ kip}\cdot\text{ft}$$

$$E_v = 5 \text{ kip}\cdot\text{ft}$$

Check governing allowable stress design combinations.

From ASCE 7-16, section 2.4, page 8

$$(D+F)+R = 40+75+10 = 125$$

From ASCE 7-16, section 2.4.5, page 9

$$1.0(D+F)+0.7E_h+0.7E_v = 40+75+0.7*25+0.7*5 = 136 \text{ kip}\cdot\text{ft}$$

$$1.0(D+F)+0.525E_h+0.525E_v+0.75S = 40+75+0.525*25+0.525*5+0.75*8.35 = 136.75 \text{ kip}$$

$$\approx 137 \text{ kip}\cdot\text{ft}$$

15. Problem

Correct Answer: (B)

Explanation:

The shear force diagram is linear, so the load must be uniformly distributed.

This rules out option D.

Since the slope of shear force does not change abruptly, there is no concentrated loading in the span. This rules out option A

There is an abrupt decrease in bending moment at midspan so the acting moment must be in anticlockwise direction. Hence the correct answer is B.

16. Problem

Correct Answer: (A)

Explanation:

Calculate the reactions from the load

Let H_1 and H_2 be horizontal reactions acting in opposite directions as shown.

Let V_1 and V_2 be vertical reactions acting as shown

Sum of forces in horizontal direction = 0

$$H_1 + 50 \text{ kip} - 50 \text{ kip} - H_2 = 0$$

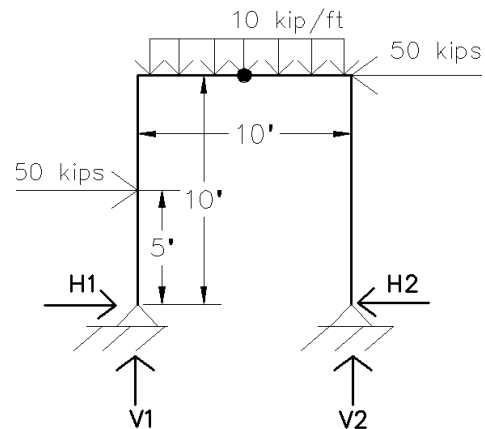
$$H_1 = H_2$$

Taking sum of moments about left pinned support

(anti clockwise +ve)

$$V_2 * 10 \text{ ft} + 50 \text{ kip} * 10 \text{ ft} - 10 \text{ kip/ft} * 10 \text{ ft} * 5 \text{ ft} - 50 \text{ kip} * 5 \text{ ft} = 0$$

$$V_2 = 25 \text{ kip}$$



Taking sum of moments at the hinge = 0 and considering forces from the right side of the hinge,

$$V_2 \cdot 5\text{ft} - H_2 \cdot 10\text{ft} - 10\text{kip/ft} \cdot 5\text{ft} \cdot 5\text{ft} / 2 = 0$$

$$25\text{kip} \cdot 5\text{ft} + H_2 \cdot 10\text{ft} - 50\text{kip} \cdot 2.5\text{ft} = 0$$

This gives $H_2 = 0$

Hence both H_1 and $H_2 = 0$.

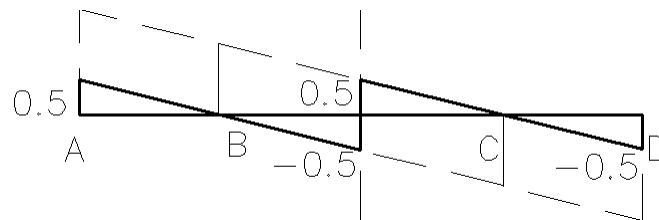
Since both horizontal reactions are zero, the only diagram that shows no shear at the reaction points is the correct answer, i.e. option A.

17. Problem

Correct Answer: (C)

Explanation:

For shear at midspan of a simply supported beam, the maximum shear will occur when the load is directly on top of midspan at either side of the section and will be zero when the load is directly on top of supports. The overhang is then extrapolated on either side as shown below. The value of the sum of ordinates at midspan will be equal to 1. Thus the influence line diagram is:



The correct answer is Option C.

18. Problem

Correct Answer: (B) Mid span at lowest sag point

Explanation:

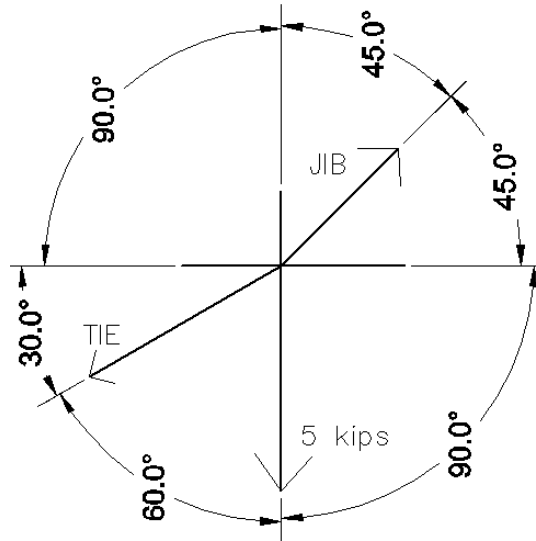
When a cable is subjected to uniformly distributed loads, it assumes a parabolic shape. The lowest point of the cable will have zero slope and zero vertical component of tension. Hence the tension is minimum at that point. Correct answer is option B.

19. Problem

Correct Answer: (C) 13.66 kips

Explanation:

The forces in the tie are tensile in nature and the forces in the jib are compressive. We can idealize the tip of the jib as a single point in equilibrium with three concurrent forces. Then from Lami's theorem, the force in any one member must be directly proportional to the sine of angle between other two forces



$$\frac{F_{jib}}{\sin(60)} = \frac{F_{tie}}{\sin(45+90)} = \frac{5 \text{ kips}}{\sin(30+90+45)}$$

$$\frac{F_{tie}}{\sin(135)} = \frac{5 \text{ kips}}{\sin(165)}$$

$$\frac{F_{tie}}{1} = \frac{5 \text{ kips} \cdot \sin(135)}{\sin(165)} = 13.66 \text{ kips}$$

Correct answer is option C.

20. Problem

Correct Answer: (C) 70.7 kips

Explanation:

Angle of inclination of braces $\Theta = \tan^{-1}(10/10) = 45^\circ$

The frame is pin connected and braced. It can be analyzed as a truss.

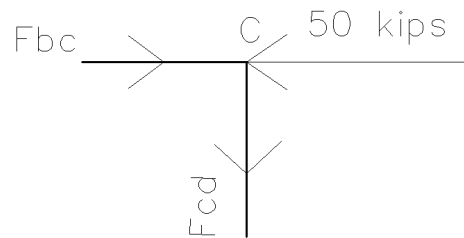
The threaded tie rods can only take tension. Hence at the given direction of loading, member AC is not engaged and can be ignored for analysis.

Analyze joint C,

BC is the only horizontal member.

Taking sum of horizontal forces = 0, we get

Force in BC = 50 kips compressive



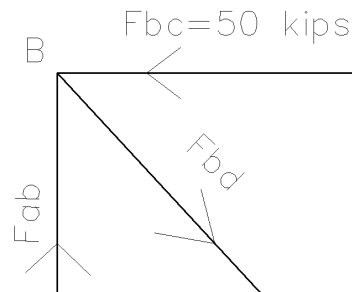
Analyze joint B of 60 kips

Taking sum of horizontal forces = 0

$F_{bd} \cos(45) = F_{bc} = 50$ kips

$F_{bd} = 50 \text{ kips} / 0.707 = 70.7$ kips

Hence the correct answer is option C.



21. Problem

Correct Answer: (D)

Explanation:

Short beams and columns are more susceptible to failure in shear. Long beams and columns usually fail in flexure and buckling.

In short beam shear stress is more concentrated due to shorter span to depth ratio.

Sudden brittle failure of concrete occurs in over reinforced sections before the reinforcement yields.

The critical value of shear in a beam is not always at a distance 'd'. E.g. if there is a concentrated load acting very close to the support, this doesn't apply.

Hence all options are true except Option D and is the correct answer.

22. Problem

Correct Answer: (B) 15.15 kips

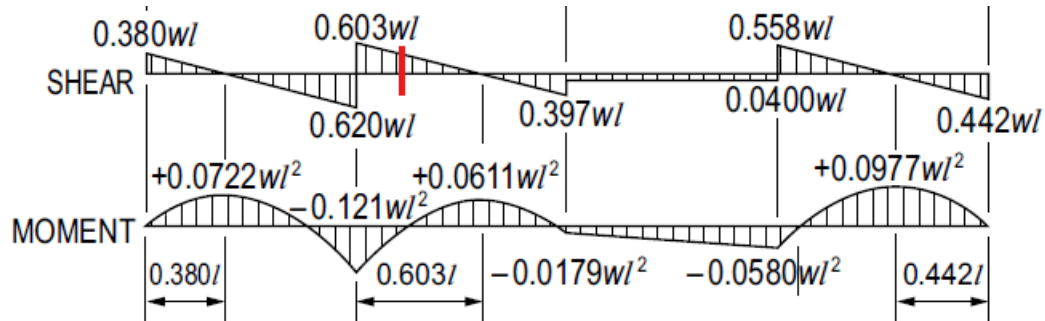
Explanation:

Ref to NCEES PE handbook page 254 for shear and bending moments in continuous beams.

Loading, $w = 5$ kips/ft

Individual span length, $L = 10$ ft

For a four span continuous system with third span not loaded, the shear force distribution is



At 13 ft from the exterior left support, the shear force varies from $0.603 \cdot w \cdot L$ to zero at the second span, $0.603 \cdot L$ from the second support.

$$0.603 \cdot w \cdot L = 0.603 \cdot 5 \text{ kips/ft} \cdot 10 \text{ ft} = 30.15 \text{ kips}$$

$$0.603 \cdot L = 0.603 \cdot 10 \text{ ft} = 6.03 \text{ ft}$$

Calculate the shear force at 13 ft i.e. at 3 ft from the second support,

$$V = 30.15 \text{ kips} / 6.03 \text{ ft} \cdot (6.03 \text{ ft} - 3 \text{ ft})$$

$V = \mathbf{15.15 \text{ kips}}$, Thus correct answer is option B.

23. Problem

Correct Answer: (B) 1604 lbs/ft

Explanation:

Calculate the maximum shear due to point loads.

Reaction at pinned end (Taking moments about roller = 0)

$$R_a \cdot (4 \text{ ft} + 6 \text{ ft} + 8 \text{ ft}) = 15 \text{ kips} \cdot 8 \text{ ft} + 10 \text{ kips} \cdot (6 \text{ ft} + 8 \text{ ft})$$

$$R_a = 14.44 \text{ kips}$$

Reaction at roller end

Taking sum of vertical forces, $\sum V = 0$

$$R_b = 10 \text{ kips} + 15 \text{ kips} - R_a$$

$$R_b = 25 \text{ kips} - 14.44 \text{ kips} = 10.56 \text{ kips}$$

Thus maximum shear occurs at R_a and the value is 14.44 kips

Shear due to uniformly distributed load $w = wL/2$

Thus, $w = 14.44 \text{ kips} \cdot 2 / (8\text{ft} + 6\text{ft} + 4\text{ft})$

$= 1.604 \text{ kips/ft}$

$= 1604 \text{ lbs/ft}$

Thus, the correct answer is option B.

24. Problem

Correct Answer: (D) 13 ksi

Explanation:

Ref to continuous beam formulas in NCEES PE handbook page 255.

Span length, $L = 10 \text{ ft}$

Loading, $w = 5 \text{ kips/ft}$

Maximum value of moment is $+0.0772wL^2$ for +ve moment and $-0.107wL^2$ for -ve moment.

Hence, governing value of bending moment is

$M = -0.107 \cdot 5 \text{ kips/ft} \cdot 10\text{ft} \cdot 10\text{ft} = 53.5 \text{ kip}\cdot\text{ft}$

Elastic flexural stress in beam is given in NCEES PE handbook page 32

$\sigma = M/s$

Ref to AISC Steel manual for value of section modulus 'S' for W10X45

From page 1-29, For W10X45, $S_{xx} = 49.1 \text{ in}^3$

$\sigma = \{53.5 \text{ kip}\cdot\text{ft} \cdot 12 (\text{in/ft})\} / 49.1 \text{ in}^3$

$= 13.07 \text{ ksi} \approx 13 \text{ ksi}$

Thus the correct answer is option D.

25. Problem

Correct Answer: (D)

Explanation:

Point of inflection is a point on the elastic curve where the deflected shape changes directions and point of contraflexure is the point where the bending moment is zero and changes signs. Thus A, B and C are correct.

Option D is false and is the answer.

26. Problem

Correct Answer: (C) 38.24 kip*ft

Explanation:

Ref NCEES PE handbook page 232 for distribution factors and page 241-244 for FEMs.

E is constant and not considered in the calculation since it cancels out in numerator and denominator for calculation of distribution factor.

Calculate joint stiffness and distribution factors for joint B (units and E not shown for calculation simplification).

Ref to NCEES PE handbook page 232 section 4.1.3, rotational stiffness “k” for far end fixed condition is given by $k = 4EI/L$

If the far end is hinged instead of fixed, $k = 3EI/L$

$$K_{bc} = 4 \cdot 700 / (12 \cdot 12) = 19.444$$

$$K_{ba} = 3 \cdot 700 / (18 \cdot 12) = 9.722$$

$$K_{bd} = 3 \cdot 1100 / (9 \cdot 12) = 30.555$$

$$K = 19.444 + 9.722 + 30.555 = 59.721$$

$$DF_{bc} = 19.444 / 59.721 = 0.326$$

$$DF_{ba} = 9.722 / 59.721 = 0.163$$

$$DF_{bd} = 30.555 / 59.721 = 0.511$$

	Joint A	Joint B			Joint C	Joint D
	AB	BA	BC	BD	CB	DB
DF	0	0.163	0.326	0.511	0	0
FEM		$5 \cdot 18^2 / 8$ = 202.5	$-5 \cdot 12^2 / 12$ = -60	$-3 \cdot 6 \cdot 3 \cdot 15 / (2 \cdot 9^2)$ $-3 \cdot 3 \cdot 6 \cdot 12 / (2 \cdot 9^2)$ = -9	$5 \cdot 12^2 / 12$ = 60	
Balancing moment		-133.5				
Distribution		-21.76	-43.521	-68.218		
Carryover	0				-21.76	
Resultant	0	180.74	-103.521	-77.218	38.24	0

Hence, correct answer is option C, 38.24 kip*ft.

27. Problem

Correct Answer: (D) 25.26 ksi

Explanation:

$$\text{Area of pipe (A)} = \pi * \{(7 \text{ in})^2 - (6 \text{ in})^2\} / 4 = 10.21 \text{ in}^2$$

$$\text{Moment of inertia (I)} = \pi * \{(7 \text{ in})^4 - (6 \text{ in})^4\} / 64 = 54.24 \text{ in}^4$$

$$\text{Section modulus (Sx)} = 54.24 \text{ in}^4 / (7/2) \text{ in} = 15.5 \text{ in}^3$$

$$\text{Tensile load (P)} = 5 \text{ kips}$$

$$\text{Moment (Mx)} = 4 \text{ kips} * 8 * 12 \text{ in} = 384 \text{ kip} * \text{in}$$

$$\begin{aligned} \text{Combined stress} &= 5 \text{ kips} / 10.21 \text{ in}^2 + 384 \text{ kip} * \text{in} / 15.5 \text{ in}^3 \\ &= \mathbf{25.26 \text{ ksi}} \end{aligned}$$

Correct answer is option D.

28. Problem

Correct Answer: (C) 3.75 kip/in

Explanation:

Here, there is no information about the eccentricity of the loads.

Thus eccentricity can be neglected.

$$\text{Total length of all round welds} = 12 \text{ in} + 12 \text{ in} + 12 \text{ in} + 12 \text{ in} = 48 \text{ in}$$

$$\text{Tension (T)} = 100 \text{ kips}$$

Tension per unit length of welds

$$f_1 = T/L = 100 \text{ kips} / 48 \text{ in} = 2.08 \text{ kip/in}$$

$$\text{Horizontal load (H)} = 150 \text{ kips}$$

Shear per unit length of welds

$$f_2 = H/L = 150 \text{ kips} / 48 \text{ in} = 3.125 \text{ kip/in}$$

Resultant stress acting on the welds

$$= \sqrt{(2.08 \text{ kip/in})^2 + (3.125 \text{ kip/in})^2} = \mathbf{3.75 \text{ kip/in}} \quad \text{Correct answer is option C}$$

29. Problem

Correct Answer: (A) 0.27 ksi

Explanation:

Cross section area of wall, $A = 6 \text{ in} \times 10 \times 12 \text{ in} = 720 \text{ in}^2$

Section modulus, $S = 6 \text{ in} \times 120 \text{ in} \times 120 \text{ in} / 6 = 14400 \text{ in}^3$

Vertical load $P = 20 \text{ kip/ft} \times 10 \text{ ft} = 200 \text{ kips}$

Total moment $M, = 360 \text{ kip}\cdot\text{ft} + 50 \text{ kips} \times 6 \text{ ft}$

$= 660 \text{ kip}\cdot\text{ft}$

$= 7920 \text{ kip}\cdot\text{in}$

Tensile stress due to bending $= M/S$

$= 7920 \text{ kip}\cdot\text{in} / 14400 \text{ in}^3 = 0.55 \text{ ksi}$

Net tensile stress $= 0.55 \text{ ksi} - 200 \text{ kips} / 720 \text{ in}^2 = \mathbf{0.27 \text{ ksi}}$

Correct answer is option A.

30. Problem

Correct Answer: (D) 1.2 in

Explanation:

Ref IBC 2018 Section Table 1604.3, page 363

For combined dead and live loading, roof members not supporting ceiling, the deflection limits is

TABLE 1604.3
DEFLECTION LIMITS^{a, b, c, h, i}

CONSTRUCTION	$L \text{ or } L_r$	$S \text{ or } W^i$	$D + L^{d, g}$
Roof members: ^c			
Supporting plaster or stucco ceiling	$L/360$	$L/360$	$L/240$
Supporting nonplaster ceiling	$L/240$	$L/240$	$L/180$
Not supporting ceiling	$L/180$	$L/180$	$L/120$

$= L/120$

$= 12 \text{ ft} / 120$

$= \mathbf{1.2 \text{ in}}$

Correct answer is option D.

31. Problem

Correct Answer: (D) 0.075in

Explanation:

Ref NCEES PE handbook page 244 for deflection on cantilever beam due to uniform load.

Dimensions of base plate = 5ftX5ft = 60inX60in

Dimensions of outrigger foot = 2ftX2ft = 24inX24in

Bearing pressure per unit width of the base plate due to outrigger load

$$w = 50 \text{ kips}/(60\text{in}) = 0.833 \text{ ksi}$$

The base plate can be idealized as a cantilevered beam from the tip of the plate to the tip of the outrigger foot.

$$\text{Cantilevered length, } l = (60\text{in} - 24\text{in})/2 = 18\text{in}$$

Moment of inertia of plate section

$$= 60\text{in} \cdot (1 \text{ in})^3/12 = 5\text{in}^4$$

$$\text{Deflection} = w \cdot l^4 / 8EI$$

$$= 0.833 \text{ (kip/in)} \cdot (18 \text{ in})^4 / (8 \cdot 29000 \text{ ksi} \cdot 5\text{in}^4)$$

$$= \mathbf{0.075\text{in}} \quad \text{Correct answer is option D.}$$

32. Problem

Correct Answer: (C) 2PL/3AE

Explanation:

Ref to NCEES PE handbook page 28 for axial deflection.

The three sections of the beam are subjected to same axial load P

Deflection in the first L/3 section

$$\Delta_1 = P \cdot (L/3) / (2A \cdot E)$$

$$\Delta_1 = PL/6AE$$

Deflection in the second L/3 section

$$\Delta_2 = P \cdot (L/3) / (A \cdot E)$$

$$\Delta_2 = PL/3AE$$

Deflection of the third L/3 section

$$\Delta_3 = P \cdot (L/3) / (2A \cdot E)$$

$$\Delta_3 = PL/6AE$$

$$\text{Total deflection} = \Delta_1 + \Delta_2 + \Delta_3$$

$$= PL/6AE + PL/3AE + PL/6AE$$

$$= \mathbf{2PL/3AE} \quad \text{Correct answer is option C.}$$

33. Problem

Correct Answer: (C) 0.5 ksi

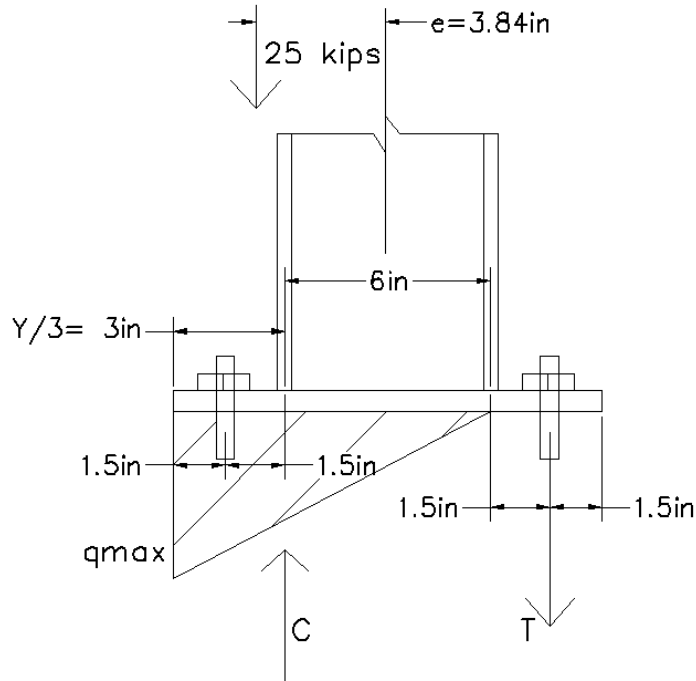
Explanation:

Calculate the eccentricity,

$$e = 8 \text{ kip}\cdot\text{ft}/25 \text{ kip}$$

$$e = 0.32 \text{ ft} = 3.84 \text{ in}$$

Eccentricity falls outside the column flange



The centroid of the compression area is at the center of the flange. Hence the length of bearing triangle, Y is given by $Y/3 = 3$ in

$$Y = 9 \text{ in}$$

Width of base plate, $B = 12$ in

To calculate compressive force ' C ', taking sum of moments about tension bolt = 0

$$C \cdot (6 \text{ in} + 1.5 \text{ in}) = 25 \text{ kips} \cdot (3.84 \text{ in} + 3 \text{ in} + 1.5 \text{ in})$$

$$C = 27.8 \text{ kips}$$

Max bearing pressure is obtained by equating area of bearing triangle to ' C '

$$0.5 \cdot q_{max} \cdot Y \cdot B = C$$

$$0.5 \cdot q_{max} \cdot 9 \text{ in} \cdot 12 \text{ in} = 27.8 \text{ kips}$$

$$q_{max} = 0.51 \text{ ksi} \approx \mathbf{0.5 \text{ ksi}}$$
 Correct answer is option C.

34. Problem

Correct Answer: (B) 30 kip

Explanation:

The collapse of the structure is due to the formation of a plastic hinge under load 'P' and fixed support A.

Let θ_1 and θ_2 be the rotations at support A and B

$$\Delta = \theta_1 * 10\text{ft}$$

$$\Delta = \theta_1 * 10\text{ft} = \theta_2 * 10\text{ft}$$

$$\theta_2 = \theta_1$$

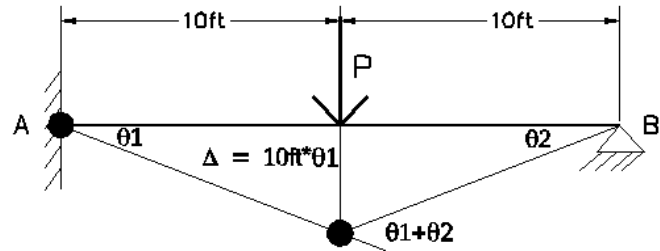
$$\text{Total rotation of plastic hinge} = \theta_1 + (\theta_1 + \theta_2) = 3\theta_1$$

External work done = internal work done

$$P * \Delta = M_p * 3\theta_1$$

$$P = M_p * 3\theta_1 / (10\text{ft} * \theta_1)$$

$$P = 3M_p / 10\text{ft} = 3 * 100\text{ kip*ft} / 10\text{ft} = \mathbf{30\text{ kip}}$$
 Correct answer is option B



35. Problem

Correct Answer: (A)

Explanation:

Ref to IBC 2018 page 417, section 1704.3.

The registered design professional in responsible charge shall prepare the statement of special inspection.

The 'engineer of record' hired by the owner is the design professional in responsible charge and is the correct answer i.e. Option A.

36. Problem

Correct Answer: (C) 1.75 inch

Explanation:

Ref to ACI 318-14, page 457, section 26.4.2

Maximum aggregate size based on thickness of slab

$$= 1/3 * \text{slab thickness} = 1/3 * 6 \text{ inch} = 2 \text{ inch}$$

Maximum aggregate size based on clear spacing of rebars

$$= 3/4 * \text{Clear spacing between rebar} = 3/4 * (3 \text{ inch} - 5/8 \text{ inch})$$

$$= 1.78 \text{ inch} \approx \mathbf{1.75 \text{ inch}}$$
 Correct answer is Option C.

37. Problem

Correct Answer: (D)

Explanation:

Option A, B and C are all the intent of construction submittals. Option D is incorrect.

Financial planning and budgeting is not based on contractor's submittals during the project.

The owner should have detailed budget plans to manage project costs before the commissioning of the project.

38. Problem

Correct Answer: (C)

Explanation:

Although all other factors influence the removal of formwork, the most important option is the compressive strength that the member has gained. The member may be cured at a very favorable temperature and may be cured with a compound which helps in rapid strength gain, but the ultimate factor that decides removal is whether it has gained sufficient strength to withstand the imposed loads. Hence option B is the correct answer.

39. Problem

Correct Answer: (B) 8 ft X 5ft

Explanation:

Per OSHA, CFR Title 29, Part 1926, Subpart L, Appendix A, (2) (a)

From table for single pole wood pole scaffolds,

For medium duty (50 psf) scaffolds up to 60 feet high,

Max pole spacing using a 4 in X 4 in = **8 ft X 5ft**

Correct answer is option B.

40. Problem

Correct Answer: (C) 27.5 ft

Explanation:

Ref to NCEES PE handbook page 153, section 3.10.3

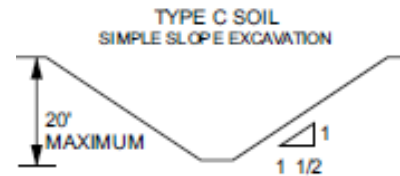
For type C soil,

Horizontal: Vertical = 1.5:1

Minimum undisturbed horizontal perimeter around the building = 25 ft / 2 = 12.5 ft

Total distance from face of building to bottom of excavated

slope = 12.5 ft + 10 ft*1.5 = **27.5 ft** Correct answer is option C



41. Problem

Correct Answer: (C) SW-SM

Explanation:

Ref to NCEES PE handbook page 115, for sieve sizes and page 110 for USCS classification.

No. 4 sieve= 4.75 mm, No. 200 sieve = 0.075mm

% of particles passing through No. 200 = 11%, thus soil is Sand or Gravel

% of particles passing No. 4 sieve = 70%, thus soil is Sand and requires dual symbol as fines (11%) are between 5% and 12%

$C_u = D_{60}/D_{10} = 0.8/0.12 = 6.67 > 6$ thus it is well graded 'SW'

LL is less than 50

PI = 46-36 = 10

From plasticity chart in page 111, it is below A line

Thus it is silty 'M' Hence the correct answer is option C, **SW-SM**.

42. Problem

Correct Answer: (D) 95%

Explanation:

Ref to NCEES PE handbook page 146 section 3.9 for standard proctor test.

Volume of mold = $1/30 \text{ ft}^3$

By inspection of the given table, maximum dry density occurs at water content 10%

$\gamma = \text{Weight} / \text{Volume}$

$= 4.35 \text{ lbf} / (1/30 \text{ ft}^3)$

$= 130.5 \text{ pcf}$

Ref NCEES PE handbook page 147

$\gamma_{Dmax} = \gamma / (1 + w)$

$\gamma_{Dmax} = 130.5 \text{ pcf} / (1 + 0.1)$

$\gamma_{Dmax} = 118.63 \text{ pcf}$

Ref NCEES PE handbook page 148, section 3.9.2

Relative compaction = dry unit weight at field/ max dry unit weight from lab test *100%

$= 112.7 \text{ pcf} / 118.63 \text{ pcf} * 100\%$

= 95%

Hence, the correct answer is Option D.

43. Problem

Correct Answer: (D) $8 \text{ ft}^3 / \text{min}$

Explanation:

Ref to NCEES PE handbook page 187 section 3.16 for Darcy's law

Permeability, $k = 0.004 \text{ ft/min}$

Hydraulic gradient, $i = \Delta h / \Delta S$

$i = (65 \text{ ft} - 45 \text{ ft}) / 90 \text{ ft}$

$I = 0.2222$

Area of flow = $30 \text{ ft} * 300 \text{ ft}$

$Q = k * i * A$

$Q = 0.004 \text{ ft/min} * 0.2222 * (30 \text{ ft} * 300 \text{ ft})$

$Q = 8 \text{ ft}^3 / \text{min}$

Correct answer is option D.

44. Problem

Correct Answer: (B) 2.5 in

Explanation:

Ref to ACI 318-14, page 336 Table 20.6.1.3.3,

Tensile strength in bending is given by the modulus of rupture

For members except walls, exposed to weather with bars between number 6 through number 11, specified concrete cover = 1.5 inches

From page 334, R20.6.1.1, specified cover should be measured to the outer edge of stirrups.

Diameter of number 4 bars = $4/8$ in = 0.5 in

Diameter of number 8 bars = $8/8$ in = 1 in

Half the diameter of number 8 bars = 0.5 in

Clear cover = 1.5 in

$c = 0.5 \text{ in} + 0.5 \text{ in} + 1.5 \text{ in} = \mathbf{2.5 \text{ in}}$

Correct answer is option B.

45. Problem

Correct Answer: (B) 2925 psi

Explanation:

28 day design strength of concrete = 6500 psi

Ref to ACI 318-14 page 409, Table 24.5.4.1

For sustained loading, allowable compressive stress

$= 0.45 * f'_c$

$= 0.45 * 6500 \text{ psi}$

$= \mathbf{2925 \text{ psi}}$

Correct answer is option B.

46. Problem

Correct Answer: (B)

Explanation:

All options are true except option B. Strain hardening increases the hardness and reduces the ductility, however, strain hardening increases the yield strength of the material. Hence option B is false and the answer.

47. Problem

Correct Answer: (A) 32.5 ksi

Explanation:

Per AISC Steel manual 15th edition, AISC 360-16, page 16.1-338

Elastic buckling stress is given by equation C-F9-1

$$b/t = 36 \text{ in} / 1 \text{ in} = 36$$

$$E = 29000 \text{ ksi}$$

$$\text{Poisson's ratio, } \nu = 0.3$$

From page 16.1-339, plate buckling coefficient $k = 1.61$

$$F_{cr} = \pi^2 * E * k / [12 * (1 - \nu^2) * (b/t)^2]$$

$$F_{cr} = \pi^2 * 29000 \text{ ksi} * 1.61 / [12 * (1 - 0.3^2) * (36)^2]$$

$$F_{cr} = \mathbf{32.5 \text{ ksi}}$$
 Correct answer is option A.

48. Problem

Correct Answer: (C) 40 kips

Explanation:

Ref to AISC steel manual page 16.1-134, Section J3.8 for slip critical connections.

From page 16.1-135, for class A surface, $\mu = 0.3$

Pretension, $T_b = 102 \text{ kips}$

Filler factor, $h_f = 1.0$

Here, factor $D_u = 1$ because T_b during installation is directly given.

Number of slip planes, $n_s = 2$ for double shear.

From equation J3-4

$$R_n = \mu * D_u * h_f * T_b * n_s$$

$$R_n = 0.3 * 1 * 1 * 102 \text{ kips} * 2 = 61.2 \text{ kips}$$

$\Omega = 1.5$ for standard size holes

$$\text{Allowable shear} = R_n / \Omega$$

$$= 61.2 \text{ kips} / 1.5 = 40.8 \text{ kips}$$

Correct answer is option C, **40 kips**.

49. Problem

Correct Answer: (B) 675 psi

Explanation:

Moisture exceeding 19% is wet condition for lumber.

Ref to NDS 2018 Supplement page 40 Table 4B

For Southern pine No. 2, 6 in wide, $F_t = 600$ psi (All adjustment values below are for F_t)

From NDS 2018 page 11, Table 2.3.2, For construction load, $C_d = 1.25$

From NDS 2018 page 11, Table 2.3.3, For 95°F, $C_t = 1$

From NDS supplement page 39, $C_m = 1$ (For F_t)

From NDS supplement page 39 for 6 in wide lumber, no size factor adjustment required,
 $C_f = 1$

From NDS 2018 page 29, Table 4.3.1,

$$F'_t = F_t * C_d * C_m * C_t * C_f * C_i$$

$$F'_t = 600 \text{ psi} * 1.25 * 1 * 1 * 1 * 0.9 = \mathbf{675 \text{ psi}} \quad \text{Correct answer is option B.}$$

50. Problem

Correct Answer: (B) 10 kips

Explanation:

Shear depth, $d_v = 32$ in

Spacing, $s = 10$ in

From TMS 402/602-16, page C-64 Table CC-6.1.2,

Area of #4 bar, $A_v = 0.2 \text{ in}^2$

From TMS 402/602-16, page C-114, section 8.3.3.1,

Allowable stress in grade 60 rebar

$$f_s = 32000 \text{ psi}$$

From TMS 402/602-16, page C-118, equation 8-27,

Shear stress carried by rebar

$$f_{vs} = 0.5 * [(A_v * f_s * d_v) / (A_{nv} * s)]$$

Rearranging,

$$f_{vs} * A_{nv} = 0.5 * [(A_v * f_s * d_v) / s]$$

$$\text{Shear force} = 0.5 * (0.2 \text{ in}^2 * 32000 \text{ psi} * 32 \text{ in}) / 10 \text{ in}$$

$$\text{Shear force} = 10240 \text{ lbs} \approx \mathbf{10 \text{ kips}} \quad \text{Correct answer is option B.}$$

51. Problem

Correct Answer: (C) 29.5 in

Explanation:

Per ACI 318-14, page 469, Table 26.6.2.1 (a)

For beams with 'd' > 8, the tolerance on 'd' is ± 0.5 in

Hence minimum tolerable dimension is 30 in - 0.5 in = **29.5 in**

Correct answer is Option C.

52. Problem

Correct Answer: (A) I and II

Explanation:

Per AISC steel manual 15th edition, RCSC specification page 16.2-8, ASTM specifications do not permit the galvanization of A490 bolts. Hence, coating test is not required. Both tensile strength test and proof load test is required. Hence, the correct answer is Option A, **I and II**.

53. Problem

Correct Answer: (C) HSS 20X20X3/4

Explanation:

Per IBC 2018 page 363, Table 1604.3, For floor beams subjected to live loads, the deflection limit is L/360 and beam subjected to dead plus live load, the deflection limit is L/240

$$L = 30\text{ft} = 360\text{ in}$$

Case 1, Live only

$$w = 5\text{ kip/ft} = 0.417\text{ kip/in}$$

$$\Delta_1 = 360\text{in} / 360$$

$$\Delta_1 = 1\text{ in}$$

From AISC steel manual 15th Ed, page 3-208, Table 3-23 case 1,

$$\Delta_1 = 5 * w * L^4 / (384 * E * I)$$

$$I = 5 * w * L^4 / (384 * E * \Delta_1)$$

$$I = 5 * 0.417\text{ kip/in} * (360\text{ in})^4 / (384 * 29000\text{ ksi} * 1\text{ in})$$

$$I = 3144.76\text{ in}^4$$

Case 2, Dead + Live

$$w = 5 \text{ kip/ft} + 1 \text{ kip/ft} = 0.5 \text{ kip/in}$$

$$\Delta 2 = 360 \text{ in} / 240$$

$$\Delta 2 = 1.5 \text{ in}$$

$$\Delta 2 = 5 * w * L^4 / (384 * E * I)$$

$$I = 5 * w * L^4 / (384 * E * \Delta 2)$$

$$I = 5 * 0.5 \text{ kip/in} * (360 \text{ in})^4 / (384 * 29000 \text{ ksi} * 1.5 \text{ in})$$

$$I = 2513.8 \text{ in}^4$$

Hence governing I is for case 1 = 3144.76 in⁴

Ref to AISC steel manual page 1-96, Table 1-12

The lightest section that satisfies the requirement is

HSS 20 X 20 X 3/4 with I = 3230 in⁴

Hence, the correct answer is Option C.

HSS20x20x7/8	0.814	221.06	60.8	21.6	21.6	3670
x3/4	0.698	191.58	52.6	25.6	25.6	3230
x5/8	0.581	161.40	44.3	31.5	31.5	2750
x1/2	0.465	130.52	35.8	40.0	40.0	2260

54. Problem

Correct Answer: (B) 910 kip*ft

Explanation:

Ref to AISC steel manual 15th ed., page 1-26 and 1-27, Table 1-1

For W12X58, Overall depth d = 12.2 in

Section modulus of W12X58, Z_x = 86.4 in³

Total depth of built up section including top and bottom 10 in X 1 in thick plates

$$D = d + 1 \text{ in} + 1 \text{ in} = 12.2 \text{ in} + 1 \text{ in} + 1 \text{ in} = 14.2 \text{ in}$$

Distance to centroid of symmetric section from plate centroid (y) = 14.2 in / 2 - 1 in / 2 = 6.6 in

Total plastic section modulus, Z_{xx} = Z_x of rolled section + 2 * Area of plates * y

$$Z_{xx} = 86.4 \text{ in}^3 + 2 * 10 \text{ in} * 1 \text{ in} * 6.6 \text{ in} = 218.4 \text{ in}^3$$

Nominal plastic moment of resistance = Z_{xx} * F_y

$$= 218.4 \text{ in}^3 * 50 \text{ ksi} = 10920 \text{ kip*in} = \mathbf{910 \text{ kip*ft}}$$

Correct answer is Option B.

55. Problem

Correct Answer: (B) 70 kips

Explanation:

$$Pr = 20 \text{ kips}$$

Required Strength Available Strength

$$Mr = 220 \text{ kip}\cdot\text{in} \qquad Mc = 500 \text{ kip}\cdot\text{in}$$

$$Vr = 4 \text{ kips} \qquad Vc = 20 \text{ kips}$$

$$Tr = 45 \text{ kip}\cdot\text{in} \qquad Tc = 200 \text{ kip}\cdot\text{in}$$

Ref to AISC steel manual 15th ed. Part ASIC360-16, page 16.1-83, section H3.2

$$Tr/Ta = 45 \text{ kip}\cdot\text{in}/200 \text{ kip}\cdot\text{in} = 22.5\%$$

Hence the interaction is given by equation H3-6

$$(Pr/Pc + Mr/Mc) + (Vr/Vc + Tr/Tc)^2 \leq 1$$

Check with $Pc = 50 \text{ kips}$

$$(20/50 + 220/500) + (4/20 + 45/200)^2 = 1.02 > 1, \text{ not good}$$

Check with $Pc = 70 \text{ kips}$

$$(20/70 + 220/500) + (4/20 + 45/200)^2 = 0.906 \text{ O.K.}$$

Hence correct answer is **70 kips**, i.e. Option B.

56. Problem

Correct Answer: (B) 2X4

Explanation:

From NDS-2018, page 11, Table 2.3.2

For Live, load, $Cd = 1$

For normal temperature and moisture conditions, $Cm = Ct = 1$

From NDS-2018, page 29, Table 4.3.1

$$F'b = Fb * Cd * Cm * Ct * CL * Cf * Cfu * Ci * Cr$$

From NDS-2018 supplement page 34, Table 4A

For DFL, select structural, $Fb = 1500 \text{ psi}$

Choose 2X4.

From NDS -2018 Supplement, page 32, $Cf = 1.5$ for 2X4

$Cfu = 1$, because it is not said that beam is loaded on flat face

Since no information on spacing is given, $Cr = 1$

$$F'b = 1500 \text{ psi} * 1 * 1 * 1 * 1 * 1.5 * 1 * 1 * 1 = 2250 \text{ psi}$$

$$\text{Allowable moment} = F'b * (b*d^2/6) = 2250 \text{ psi} * (2 \text{ in} * 4 \text{ in} * 4 \text{ in} / 6)$$

$$= 12000 \text{ lb-in} = 1000 \text{ lb-ft} . \text{ Hence it is adequate}$$

Correct answer is Option B.

57. Problem

Correct Answer: (B) #5

Explanation:

From NCEES PE handbook page 260, section 4.3.1,

$$\text{Area of \#4 bars} = 0.2 \text{ in}^2$$

$$\text{Area of \#5 bars} = 0.31 \text{ in}^2$$

Try with five #4 bars

$$\text{Total } A_s = 5 * 0.2 \text{ in}^2 = 1 \text{ in}^2$$

From NCEES PE handbook page 264, section 4.3.2.2

$$a = (A_s * f_y) / (0.85 * f'_c * b)$$

$$a = (1 \text{ in}^2 * 60 \text{ ksi}) / (0.85 * 4 \text{ ksi} * 9 \text{ in})$$

$$a = 1.961 \text{ in}$$

$$M_n = A_s * f_y * (d - a/2)$$

$$M_n = 1 \text{ in}^2 * 60 \text{ ksi} * (18 \text{ in} - 1.961 \text{ in} / 2)$$

$$M_n = 1021.17 \text{ kip-in} = 85.1 \text{ kip-ft (not enough)}$$

Try five #5 bars

$$A_s = 5 * 0.31 \text{ in}^2 = 1.55 \text{ in}^2$$

$$a = (1.55 \text{ in}^2 * 60 \text{ ksi}) / (0.85 * 4 \text{ ksi} * 9 \text{ in})$$

$$a = 3.039 \text{ in}$$

$$M_n = 1.55 \text{ in}^2 * 60 \text{ ksi} * (18 \text{ in} - 3.039 \text{ in} / 2)$$

$$M_n = 1532.68 \text{ kip-in} = 127.7 \text{ kip-ft}$$

From ACI 318-14, page 343, Table 21.2.2 for tension controlled section,

$$\Phi = 0.9$$

$$\text{Design strength} = \Phi * M_n$$

$$= 0.9 * 127.7 \text{ kip-ft}$$

$$= 114.9 \text{ kip-ft}$$

Hence #5 bars satisfies the requirement and the correct answer is Option B.

58. Problem

Correct Answer: (A) 6in

Explanation:

From NCEES PE handbook page 260, section 4.3.1,

Area of #3 bars = 0.11 in^2

For 2 legged #3 stirrups, $A_v = 2 * 0.11 \text{ in}^2 = 0.22 \text{ in}^2$

From ACI 318-14, page 341 Table 21.2.1, strength reduction factor for shear

$\Phi = 0.75$

Required Shear strength, $V_u = 51 \text{ kips}$

From ACI 318-14, page 316, Table 19.2.4.2, for normal weight concrete, $\lambda = 1$

From ACI 318-14, page 354, section 22.5.5.1,

Shear capacity provided by concrete

$$V_c = 2\lambda(\sqrt{f'_c}) * b_w * d$$

$$V_c = 2 * 1 * (\sqrt{4000}) * 12 \text{ in} * 18 \text{ in}$$

$$V_c = 27322.1 \text{ lbs} = 27.3 \text{ kips}$$

$V_u > \Phi V_c$, so stirrups are required

From ACI 318-14, page 359, section 22.5.10

$$V_s = V_u / \Phi - V_c$$

$$V_s = 51 \text{ kips} / 0.75 - 27.3 \text{ kips}$$

$$V_s = 40.7 \text{ kips}$$

From ACI 318-14, page 359, equation 22.5.10.5.3

$$V_s = A_v * f_y * d / s$$

$$s = A_v * f_y * d / V_s$$

$$s = 0.22 \text{ in}^2 * 60 \text{ ksi} * 18 \text{ in} / 40.7 \text{ kips}$$

$$s = 5.8 \text{ in}$$

Hence, the correct answer is Option A, **6in**.

59. Problem

Correct Answer: (C) 82.8%

Explanation:

$$Kl_x = Kl_y = 12 \text{ ft}$$

From AISC steel manual page 4-24, table 4-1 a

For W8X40 with $L_c = 12 \text{ ft}$,

$$P_n/\Omega = P_c = 243 \text{ kips}$$

$$P_r/P_c = 100 \text{ kips} / 243 \text{ kips} = 0.412 > 0.2$$

For flexure, lateral torsional buckling is prevented, so yield strength is verified.

From AISC Steel manual page 3-27, table 3-2

For W-8X40

$$M_{px}/\Omega = 99.3 \text{ kip}\cdot\text{ft} = M_{cx}$$

From AISC Steel manual page 3-32, table 3-4

For W-8X40

$$M_{ny}/\Omega = 46.2 \text{ kip}\cdot\text{ft} = M_{cy}$$

From steel manual page 16.1-77, equation H1-1a

$$P_r/P_c + (8/9) * (M_{rx}/M_{cx} + M_{ry}/M_{cy})$$

$$= 100 \text{ kips} / 243 \text{ kips} + (8/9) * (25 \text{ kip}\cdot\text{ft} / 99.3 \text{ kip}\cdot\text{ft} + 10 \text{ kip}\cdot\text{ft} / 46.2 \text{ kip}\cdot\text{ft})$$

$$= \mathbf{82.8\%} \quad \text{Hence the correct answer is Option C.}$$

60. Problem

Correct Answer: (B) 20 ksi

Explanation:

From AISC steel manual page 15th ed. Page 1-118, Table 1-16

For 2C12X30 spaced at 3/8in, $r_y = 1.15 \text{ in}$

$$KL/r_y = 10 \text{ ft} * 12 \text{ (in/ft)} / 1.15 \text{ in}$$

$$KL/r_y = 104.34$$

From steel manual page 4-231, Table 4-14,

For 36 ksi steel and $L/r = 105$

$$F_{cr}/\Omega = 12.1 \text{ ksi}$$

$$F_{cr} = 12.1 \text{ ksi} * \Omega$$

$$F_{cr} = 12.1 \text{ ksi} * 1.67$$

$$F_{cr} = 20.2 \text{ ksi} \quad \text{Hence the correct answer is Option B, } \mathbf{20 \text{ ksi}.}$$

61. Problem

Correct Answer: (B)

Explanation:

Per TMS 402/602-16 page C-56 and C-57, section 5.3.1

Minimum vertical reinforcement should be $0.0025A_n$

For 16 in X 16 in column $A_{smin} = 0.0025 * 16 \text{ in} * 16 \text{ in} = 0.64 \text{ in}^2$.

Option A is not true.

Minimum side dimension should be 8 in, so Option C is not true.

Per TMS 402/602-16 page C-57, section 5.3.1.4 (b)

Tie spacing should not exceed $48 * \text{tie diameter}$

For #4 tie, spacing = $48 * 4 / 8 \text{ in} = 24 \text{ in}$. So Option D is not true

From section 5.3.1.1 (a), $h < 99 * r$

$h < 99 * 4 \text{ in}$

$h < 33 \text{ ft}$

For cantilevered column, from figure CC-5.3-1, $h = 2 * \text{height}$

For 17 ft cantilevered column, $h = 17 \text{ ft} * 2 = 34 \text{ ft}$ i.e. h exceeds 33ft and is not permitted

Hence Option B is true and the answer.

62. Problem

Correct Answer: (A) 339 psi

Explanation:

From NDS-2018 supplement page 14, Table 1B, standard dressed size for $2X8 = 1.5X7.25$

From NDS-2018 supplement page 34, Table 4A, for DFL No. 1 & Btr,

$E_{min} = 660000 \text{ psi}$

For normal temperature and dry conditions, $C_m = C_t = 1$

From NDS-2018, page 29, Table 4.3.1

$E_{min}' = E_{min} * C_m * C_t * C_i * C_T$

$E_{min}' = 660000 \text{ psi} * 1 * 1 * 1 * 1 = 660000 \text{ psi}$

Planks are pin connected, effective length factor = 1

$L_x/d_1 = 10 \text{ ft} / 7.25 \text{ in} = 120 \text{ in} / 7.25 \text{ in} = 16.55$

$L_y/d_2 = 5 \text{ ft} / 1.5 \text{ in} = 60 \text{ in} / 1.5 \text{ in} = 40$ (L_y governs)

From NDS-2018, page 21, section 3.7.1, critical buckling stress is given by

$$F_{cE} = 0.822 * E_{min}' / (l_e/d)^2$$

$$F_{cE} = 0.822 * 660000 \text{ psi} / (40)^2 = 339.07 \text{ psi}$$

Hence the correct answer is Option A, **339 psi**.

63. Problem

Correct Answer: (B) **12 in**

Explanation:

Ref to ACI 318-14, page 507, Appendix A for bar size diameters

Per ACI 318-14, page 438, section 25.7.2.1 (b)

Center-to-center spacing shall not exceed the least of 16*db of longitudinal bar, 48*db of tie bar, and smallest dimension of member

- 1) For #7 longitudinal bar, $16*db = 16*7/8 \text{ in} = 14 \text{ in}$
- 2) For #3 Tie bar, $48*3/8 \text{ in} = 18 \text{ in}$
- 3) Minimum dimension of the member = 12in

Hence **12 in** governs and the correct answer is Option B.

64. Problem

Correct Answer: (B) Vertical bracing

Explanation:

In braced frames, the diaphragms assist in distributing the loads efficiently from the floors and roof to the bracing elements. However, the primary element that contributes to lateral story stiffness, strength and stability in a braced frame are the vertical bracings. Moment connections are typically not designed in braced frames and could only serve as redundant elements. Foundation anchors help transfer the superstructure load to the substructure and is not the answer. Hence, Option B is the correct answer.

65. Problem

Correct Answer: (C) 9.57 kips

Explanation:

From AISC Steel manual 15th edition, page 16.1-105, section 2a

When steel deck ribs are perpendicular to the beam and distance between edge of stud shank and deck web is > 2 inches, $R_p = 0.75$ (strong position)

From AISC steel manual, page 3-203, Table 3-21,

For $F_u = 65$ ksi, Normal weight 4ksi concrete and deck perpendicular with $R_p = 0.75$

For 1 stud per rib, 0.5 inch diameter stud, $Q_n = 9.57$ kips

Correct answer is Option C.

66. Problem

Correct Answer: (D) 8

Explanation:

Length of channel, $l_a = 6$ in

$f'_c = 4$ ksi

$E = 3600$ ksi

From AISC steel manual 15th ed., page 1-38, Table 1-5,

For C 3X6,

$t_w = 0.356$ in

$t_f = 0.273$ in

From AISC steel manual 15th ed., page 16.1-106, eqn. I8-2,

$$Q_n = 0.3 \cdot (t_f + 0.5 \cdot t_w) \cdot l_a \cdot \sqrt{f'_c \cdot E}$$

$$Q_n = 0.3 \cdot (0.273 \text{ in} + 0.5 \cdot 0.356 \text{ in}) \cdot 6 \text{ in} \cdot \sqrt{4 \text{ ksi} \cdot 3600 \text{ ksi}}$$

$$Q_n = 97.4 \text{ kips}$$

Total force to be transferred = 380 kips.

The point of max moment to zero moment in a simply supported beam with Uniformly distributed load is half the span length,

$$\text{Total number of channels required over the full span} = 2 \cdot 380 \text{ kips} / Q_n$$

$$= 2 \cdot 380 \text{ kips} / 97.4 \text{ kips} = 7.8$$

Hence total number of channels required = 8, correct answer is Option D

67. Problem

Correct Answer: (B) 7.1 ksi

Explanation:

For an unshored construction, the entire load is taken by the beam. Immediately after pouring, the concrete won't have hardened and there is no composite action. All the load is taken by the steel beam. The given section modulus in the question is of no use.

Also, before the concrete has hardened, there won't be any live load application so only dead load is considered.

From AISC steel manual 15th Ed. page 1-22 and 1-23, Table 1-1,

For W18X71, $S_x = 127 \text{ in}^3$

Factored bending stress = Factored Moment / Section modulus

$$= 75 \text{ kip}\cdot\text{ft} \times 12 \text{ (in/ft)} / 127 \text{ in}^3 = \mathbf{7.1 \text{ ksi}}$$

Hence, the correct answer is Option B.

68. Problem

Correct Answer: (A) 33 kips

Explanation:

From AISC steel manual 15th ed. , page 16.1-116, Section J1.8 (d)

The high-strength bolts shall have an available strength of not less than 33% of the required strength of the connection.

$$33/100 \times 100 \text{ kips} = \mathbf{33 \text{ kips}}$$

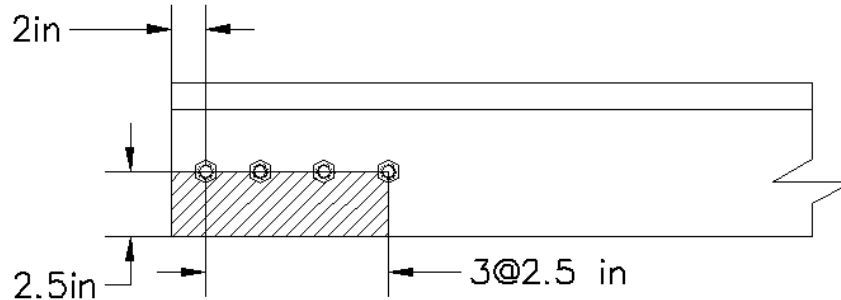
Correct answer is Option A.

69. Problem

Correct Answer: (B) 81 kips

Explanation:

The shaded area represents area that splits from the member during block shear failure



From AISC steel manual 15th ed. , page 16.1-130, Table J3.3, nominal hole size for standard holes with 3/4 in. diameter bolts = 13/16 in.

For net area calculation per page 16.1-20, section B4.3b, width of a bolt hole shall be taken as 1/16 in. greater than the nominal dimension of the hole,

$$d = 13/16 \text{ in} + 1/16 \text{ in} = 0.875 \text{ in}$$

Net tension area, A_{nt}

$$\begin{aligned} &= (2.5 \text{ in} - 0.875 \text{ in} / 2) * t \\ &= 2.06 \text{ in} * 0.5 \text{ in} = 1.03 \text{ in}^2 \end{aligned}$$

Net shear area, A_{nv}

$$= (3 * 2.5 \text{ in} + 2 \text{ in} - 3.5 * 0.875 \text{ in}) * 0.5 \text{ in} = 3.21 \text{ in}^2$$

Gross shear area, A_{gv}

$$= (3 * 2.5 \text{ in} + 2 \text{ in}) * 0.5 \text{ in} = 4.75 \text{ in}^2$$

From AISC steel manual 15th ed. Page 16.1-138, equation J4-5,

$$R_n = 0.6 * F_u * A_{nv} + U_{bs} * F_u * A_{nt} \quad (U_{bs} = 1)$$

$$\begin{aligned} R_n &= 0.6 * 58 \text{ ksi} * 3.21 \text{ in}^2 + 1 * 58 \text{ ksi} * 1.03 \text{ in}^2 \\ &= 171.45 \text{ kips} \end{aligned}$$

Check $0.6 * F_y * A_{gv} + U_{bs} * F_u * A_{nt}$

$$\begin{aligned} &= 0.6 * 36 \text{ ksi} * 4.75 \text{ in}^2 + 1 * 58 \text{ ksi} * 1.03 \text{ in}^2 \\ &= 162.34 \text{ kips (governs)} \end{aligned}$$

Allowable block shear strength = $162.34 \text{ kips} / \Omega$

$$= 162.34 \text{ kips} / 2 \approx \mathbf{81 \text{ kips}}$$

Hence the correct answer is Option B.

70. Problem

Correct Answer: (B) **2.98 in²**

Explanation:

Per AISC steel manual 15th ed. page 16.1-20, section B4.3b, width of a bolt hole shall be taken as 1/16 in. greater than the nominal hole dimension, = 13/16in + 1/16in = 0.875 in

From AISC steel manual 15th ed, page 1-48 and 1-49, Table 1-7, for L 5 X 3 X 1/2

Height of centroid, $x = 0.746$ in Area of section, $A_g = 3.75$ in²

Net area after subtracting bolt hole, $A_n = 3.75$ in² - 0.875 in * 0.5 in = 3.3125 in²

Length of connection, $l = 3 * 2.5$ in = 7.5 in

From AISC steel manual 15th ed, page 16.1-30, Table D3.1, Case 2

$U = 1 - x/l = 1 - 0.746$ in / 7.5 in = 0.9

From page 16.1-29, Eqn. D3-1, $A_e = A_n * U$

$A_e = 0.9 * 3.3125$ in² = **2.98 in²**. Correct answer is Option B

71. Problem

Correct Answer: (C)

Explanation:

High strength bolts and larger welds contribute to strength and not overall ductility.

Reducing the number of bolts does not necessarily increase ductility and may complicate the load distribution. Designing the connection so that yielding occurs in the connected parts (e.g., plates or angles) before bolt or weld failure ensures adequate energy dissipation ductile behavior during cyclic loadings.

Correct answer is Option C.

72. Problem

Correct Answer: (C) 6

Explanation:

The connection satisfies all requirements per AISC steel manual 15th ed. Page 10-9, 'Design table discussions'. So Table 10-1 can be directly used.

Total reaction to be transferred = Dead load + Live Load
= 20 kip + 50 kips = 70 kips

From AISC steel manual Page 10-16, Table 10-1

For 6 rows of bolts, Group A (A325) SC class A, 3/4 in diameter bolts, STD holes,

Available strength for 5/16 inch thick angle = 75.9 kips, satisfies requirement.

Hence the correct answer is Option C, 6.

73. Problem

Correct Answer: (D) 91.2 kips

Explanation:

From AISC steel manual 15th edition, page 8-92, Table 8-8,

$e_x = a \cdot l$,

i.e. $a = 4 \text{ in} / 8 \text{ in} = 0.5$

$K \cdot l = 4 \text{ in}$

$k = 4 \text{ in} / 8 \text{ in} = 0.5$

For $a = 0.5$ and $k = 0.5$, $C = 2.85$

Size of 1/2 inch weld in 16th of an inch,

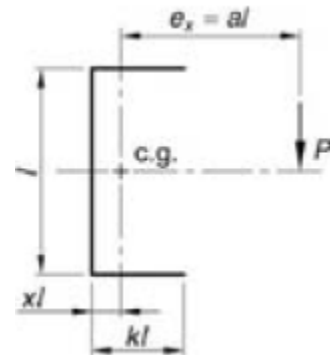
1/2 in = 8/16 in. Hence $D = 8$

From AISC steel manual 15th edition, page 8-67, Table 8-3,

$C_1 = 1$ for $F_{exx} = 70 \text{ ksi}$

Allowable Load = $C \cdot C_1 \cdot D \cdot l / \Omega$

= $2.85 \cdot 1 \cdot 8 \cdot 8 / 2 = 91.2 \text{ kips}$ Correct answer is Option D.



74. Problem

Correct Answer: (B) 39 kips

Explanation:

$F_{exx} = 70 \text{ ksi (SMAW)}$

Total length of welds (two sides of plate), $L = 2 \times 6 \text{ in} = 12 \text{ in}$

From AISC steel manual 15th edition, page 1-97, table 1-12

For HSS 8X8X1/4, design wall thickness, $t = 0.233 \text{ in}$

From AISC steel manual 15th edition, page 16.1-118, table J2.2

$R = 2 \times t = 2 \times 0.233 \text{ in} = 0.466 \text{ in}$

For flare bevel done by SMAW, effective throat thickness

$= \frac{5}{16} \times R = \frac{5}{16} \times 0.466 \text{ in} = 0.146 \text{ in}$

Throat area = effective throat thickness $\times L = 0.146 \text{ in} \times 12 \text{ in} = 1.752 \text{ in}^2$

From AISC steel manual 15th edition, page 16.1-123, table J2.5

For flare bevel in tension normal to weld axis,

Allowable strength = $0.6 \times F_{exx} \times \text{throat area} / \Omega$

Allowable strength = $0.6 \times 70 \text{ ksi} \times 1.752 \text{ in}^2 / 1.88 \approx \mathbf{39 \text{ kips}}$

Correct answer is Option B.

75. Problem

Correct Answer: (B) 250 lbs

Explanation:

From NDS 2018 page 181, Table L2, for standard hex lag screws with

Diameter, $D = 3/8 \text{ in}$ Length, $L = 7 \text{ in}$

Minimum thread length, $T = 4 \text{ in}$

Length of thread penetration in member,

$p = T - E = 3.78 \text{ in}$

From NDS 2018, page 87, Table 12.3.3A

Specific gravity for DFL, $G = 0.5$

From NDS 2018, page 77, Table 12.2A

For $G = 0.5$, using $3/8 \text{ inch}$ diameter lag screws,

Withdrawal value, $W = 305 \text{ lbs/in}$

From NDS 2018, page 106, Table 12J

For DFL side member thickness 2.5 in. and $D = \frac{3}{8}$ in

$$Z_{||} = 200 \text{ lbs}$$

$$Z_{\perp} = 140 \text{ lbs (governs)}$$

From NDS 2018, page 89, eqn. 12.4-1

$$Z_{\alpha} = W_p Z / (W_p \cos^2 \alpha + Z \sin^2 \alpha)$$

$$Z_{\alpha} = 305 \cdot 3.78 \cdot 140 / (305 \cdot 3.78 \cdot \cos^2 45^\circ + 140 \cdot \sin^2 45^\circ)$$

$$Z_{\alpha} = 249.7 \text{ lbs} \approx \mathbf{250 \text{ lbs}}$$

Correct answer is Option B.

76. Problem

Correct Answer: (C) 3ft

Explanation:

$$\text{Weight of soil, } W_s = 110 \text{ pcf} \cdot 5 \text{ ft} \cdot 25 \text{ ft} \cdot 50 \text{ ft} = 687500 \text{ lbs} = 687.5 \text{ kips}$$

$$\text{Dead load, } D = 1500 \text{ kips}$$

$$\text{Weight of Mat} = W_m$$

$$\text{Uplift} = 1800 \text{ kips}$$

$$\text{Factor of safety, } FS = 1.5$$

$$\text{Resisting weight} / \text{Uplift} = FS$$

$$W_s + D + W_m = \text{Uplift} \cdot FS$$

$$687.5 \text{ kips} + 1500 \text{ kips} + W_m = 1800 \text{ kips} \cdot 1.5$$

$$W_m = 512.5 \text{ kips}$$

$$\text{Required depth of mat} = W_m / (\text{Area} \cdot \text{unit weight})$$

$$= 512.5 \text{ kips} / (25 \text{ ft} \cdot 50 \text{ ft} \cdot 0.145 \text{ kcf})$$

$$= 2.83 \text{ ft}$$

Correct answer is Option C, **3ft**

77. Problem

Correct Answer: (D) 2.03 ksf

Explanation:

Weight of footing = $1.8 \text{ klf} \times 25 \text{ ft} = 45 \text{ kips}$

Total Vertical load, $P = 50 \text{ kips} - 15 \text{ kips} + 45 \text{ kips} = 80 \text{ kips}$

Total overturning Moment about the left edge of footing,

$M_o = 15 \text{ kip} \times (25 \text{ ft} - 2.5 \text{ ft}) = 337.5 \text{ kip}\cdot\text{ft}$

Total resisting moment about the left edge of the footing

$M_r = 50 \text{ kip} \times 2.5 \text{ ft} + 45 \text{ kips} \times 25 \text{ ft} / 2 = 687.5 \text{ kip}\cdot\text{ft}$

Location of resultant from left edge

$X = (M_r - M_o) / P = (687.5 \text{ kip}\cdot\text{ft} - 337.5 \text{ kip}\cdot\text{ft}) / 80 \text{ kips} = 4.375 \text{ ft}$

Eccentricity from center of footing, $e = B/2 - x = 25 \text{ ft} / 2 - 4.375 \text{ ft} = 8.125 \text{ ft}$

Thus $e > B/6$

From NCEES PE handbook page 93,

$Q_{\max} = 4P / [3L(B - 2e)]$

$Q_{\max} = 4 \times 80 \text{ kips} / [3 \times 6 \text{ ft} (25 \text{ ft} - 2 \times 8.125 \text{ ft})]$

$Q_{\max} = \mathbf{2.03 \text{ ksf}}$

Correct answer is Option D.

78. Problem

Correct Answer: (B) 4ft

Explanation:

From IBC 2018 page 434, Table 1806.2,

Presumptive load bearing value for GW and GP for vertical foundation pressure

$= 3000 \text{ psf} = 3 \text{ ksf}$

Pressure due to column load + footing weight + soil weight = Presumptive value

$P/B^2 + 0.15 \text{ kcf} \times 2 \text{ ft} + 0.115 \text{ kcf} \times (3 \text{ ft} - 2 \text{ ft}) = 3 \text{ ksf}$

$40 \text{ kips} / B^2 = 2.585 \text{ ksf}$

$B^2 = 40 \text{ kips} / 2.585 \text{ ksf}$

$B = 3.93 \text{ ft}$

Hence, the correct answer is Option B, **4ft**.

79. Problem

Correct Answer: (A) 10 kips

Explanation:

Scour depth = 8 ft

Hence effective length of pile in the first layer = 10 ft - 8 ft = 2 ft

First layer

$q_1 = 200 \text{ plf}$ $L_1 = 2 \text{ ft}$

Second layer

$q_2 = 300 \text{ plf}$ $L_2 = 12 \text{ ft}$

Third layer

$q_3 = 400 \text{ plf}$ $L_3 = 15 \text{ ft}$

Total frictional resistance = $q_1 * L_1 + q_2 * L_2 + q_3 * L_3$

= $200 \text{ plf} * 2 \text{ ft} + 300 \text{ plf} * 12 \text{ ft} + 400 \text{ plf} * 15 \text{ ft}$

= 10000 lbs

= **10 kips**

Correct answer is option A.

80. Problem

Correct Answer: (B) 9 ft

Explanation:

The closest inclined crack to the support of the cantilevered wall will extend upward from the face of the base reaching the compression zone approximately 'd' from the face of the base as marked in the figure with red.

The critical section of stem for shear is at a distance 'd' from the face of the base.

$d = 15 \text{ in} - \text{effective cover} = 15 \text{ in} - 3 \text{ in} = 12 \text{ in}$

Distance from the top of the wall

$D = 11.5 \text{ ft} - 18 \text{ in} - 12 \text{ in} = \mathbf{9 \text{ ft}}$

Correct answer is option B.

